MECHANICAL LEGEND

(APPLICABLE TO ALL MECHANICAL DRAWINGS)

SYMBOL	DESCRIPTION
RL	REFRIGERANT LIQUID
RS	REFRIGERANT SUCTION
A / M-1	PLAN/ SECTION DESIGNATION TOP - PLAN/ SECTION REFERENCE, BOTTOM - REFERENCED DRAWING
	UNION
	EXHAUST AIR DUCT
	OUTDOOR AIR OR SUPPLY AIR DUCT
	RETURN AIR DUCT
T	THERMOSTAT/TEMPERATURE SENSOR
S	DUCT MOUNTED SMOKE DETECTOR
D	DROP IN DUCTWORK
R	RISE IN DUCTWORK
	VOLUME DAMPER
	FLEXIBLE CONNECTION
	SINGLE DUCT AIR CONDITIONING TERMINAL UNIT

GENERAL MECHANICAL NOTES

(APPLICABLE TO ALL MECHANICAL DRAWINGS)

1. COORDINATE ALL MECHANICAL WORK WITH PLUMBING WORK, ELECTRICAL WORK, AND WORK OF OTHER TRADES SHOWN ON OTHER DRAWINGS.
2. RUN ALL SOIL, WASTE AND DRAIN PIPING WITH 2 PERCENT MINIMUM GRADE UNLESS OTHERWISE NOTED. HORIZONTAL VENT PIPING SHALL BE GRADED TO DRIP BACK TO THE SOIL OR WASTE PIPE BY GRAVITY.
3. INSTALL DUCTWORK SO THAT ALL DAMPERS ARE ACCESSIBLE.
4. UNLESS OTHERWISE NOTED, ROUTE ALL DUCTWORK OVERHEAD, TIGHT TO UNDERSIDE OF SLAB, WITH SPACE FOR INSULATION IF REQUIRED.
5. MAINTAIN MINIMUM 6'-8" CLEARANCE TO UNDERSIDE OF PIPES, DUCTS, CONDUITS, SUSPENDED EQUIPMENT, ETC., THROUGHOUT ACCESS ROUTES IN MECHANICAL AND ELECTRICAL ROOMS.
6. CERTAIN ITEMS SUCH AS ACCESS DOORS, CLEANOUTS, RISE AND DROPS IN DUCTWORK, ETC., ARE INDICATED ON THE DRAWINGS FOR CLARITY OR A SPECIFIC LOCATION REQUIREMENT AND SHALL NOT BE INTERPRETED AS THE EXTENT OF THE REQUIREMENTS FOR THESE ITEMS. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THESE ITEMS AS REQUIRED ELSEWHERE IN THE CONTRACT DOCUMENTS.
7. WHERE THE INSTALLATION OF NEW SERVICES OR THE EXTENSION OF EXISTING SERVICES REQUIRES CUTTING OF EXISTING FLOORS, WALLS, PARTITIONS, ETC., CHECK FOR THE PRESENCE OF EXISTING MECHANICAL, PLUMBING AND ELECTRICAL SERVICES WITHIN OR IMMEDIATELY BENEATH CONSTRUCTION. EXERCISE NECESSARY PRECAUTIONS TO PREVENT DAMAGE TO THE SERVICES OR INJURY TO PERSONNEL DUE TO CONTACT WITH SAME. TEMPORARILY DISCONNECT SERVICES DURING THE CUTTING OPERATION. SCHEDULE SERVICE OUTAGES IN ADVANCE WITH THE OWNER.
8. FLOW SCHEMATIC AND RISER DIAGRAMS INDICATE FLOW AND OPERATION CONCEPTS AS WELL AS GENERAL ARRANGEMENT OF EQUIPMENT. VALVES, PRESSURE GAUGES, ETC. ARE INDICATED FOR THIS PURPOSE. PROVIDE ADDITIONAL VALVES, PRESSURE GAUGES, ETC. AS SHOWN ON VARIOUS EQUIPMENT DETAILS. SEE PLANS AND DETAILS FOR PIPE SIZES NOT INDICATED ON FLOW SCHEDULES AND RISER DIAGRAMS.
9. CONTRACTOR SHALL BE RESPONSIBLE FOR RESEARCHING ALL SYSTEMS THAT A PARTICULAR OUTAGE WILL AFFECT AS WELL AS LOCATING ALL SHUTOFF POINTS. INCLUDE THIS INFORMATION IN THE OUTAGE PLAN AND SUBMIT TO THE OWNER FOR APPROVAL.
10. EXCEPT AS OTHERWISE NOTED, LOCATE ALL ROOM THERMOSTATS ABOVE FINISHED FLOOR ON SAME HORIZONTAL CENTERLINE AS LIGHT SWITCH. WHERE LIGHT SWITCH AND [THERMOSTAT]/[TEMPERATURE SENSOR] ARE ADJACENT TO EACH OTHER, LIGHT SWITCH SHALL BE CLOSEST TO THE DOOR. COORDINATE WITH ELECTRICAL CONTRACTOR. NOTIFY THE ENGINEER OF ROOMS WHERE THE ABOVE LOCATION CANNOT BE MAINTAINED OR WHERE THERE IS A QUESTION ON LOCATION.
11. IN CORRIDORS WHERE CEILING SPEAKERS AND AIR DIFFUSERS ARE INDICATED BETWEEN THE SAME LIGHTING FIXTURES, RELOCATE BOTH DEVICES TO QUARTER POINTS BETWEEN THE SAME FIXTURE.

MECHANICAL DRAWING LIST

SHEET NUMBER	SHEET NAME
M001	MECHANICAL COVER SHEET
M100	CRAWL SPACE - MECHANICAL - NEW WORK
M101	FIRST FLOOR - MECHANICAL - NEW WORK
M102	SECOND FLOOR - MECHANICAL - NEW WORK
M103	ROOF - MECHANICAL - NEW WORK
M401	ENLARGED MECHANICAL PLANS
M501	MECHANICAL DETAILS
M601	MECHANICAL SECTIONS
M701	AUTOMATIC CONTROLS
M702	AUTOMATIC CONTROLS
M703	AUTOMATIC CONTROLS
M801	MECHANICAL SCHEDULES

MECHANICAL ABBREVIATIONS

(APPLICABLE TO ALL MECHANICAL DRAWINGS)

ACCU	AIR COOLED CONDENSING UNIT
ACU	AIR CONDITIONING UNIT
AFM	AIR FLOR MONITOR
APD	AIR PRESSURE DROP
ATU	AIR TERMINAL UNIT
BAS	BUILDING AUTOMATION SYSTEM
BDD	BACK DRAFT DAMPER
BHP	BRAKE HORSEPOWER
BTU	BRITISH THERMAL UNIT
CAP	CAPACITY
CFH	CUBIC FEET PER HOUR
CFM	CUBIC FEET PER MINUTE
CO	CARBON DIOXIDE SENSOR
CS	CURRENT SENSOR
D	DIAMETER / DIFFUSER / DEPTH / DROP
DB	DRY BULB
DN	DOWN
DPS	DIFFERENTIAL PRESSURE SWITCH
DWH	DOMESTIC WATER HEATER
EA	EXHAUST AIR
EDB	ENTERING DRY BULB
EF	EXHAUST FAN
EH	ELECTRIC HEATER
ESP	EXTERNAL STATIC PRESSURE
ESS	EMERGENCY SHUT-OFF SWITCH
EWB	ENTERING WET BULB
FLA	FULL LOAD AMPS
FV	FACE VELOCITY
H	HEIGHT
HC	HEATING COIL
HOA	HAND-OFF-AUTOMATIC
HP	HORSEPOWER
KW	KILOWATT
L	LENGTH
LDB	LEAVING DRY BULB
LS	LIGHT SWITCH
M	MECHANICAL
MAU	MAKEUP AIR UNIT
MAX	MAXIMUM
MBH	THOUSAND BTU PER HOUR
MIN	MINIMUM
N	NORTH
NO	NUMBER
OA	OUTDOOR AIR
OED	OPEN END DUCT
OS	OCCUPANCY SENSOR
PD	PRESSURE SWITCH
PH	PHASE
PRESS	PRESSURE
R	REGISTER / RISE
RA	RETURN AIR
RH	RADIANT HEATER
RL	REFRIGERANT LIQUID
RPM	REVOLUTIONS PER MINUTE
RS	REFRIGERANT SUCTION
RTU	ROOFTOP UNIT
SA	SUPPLY AIR
SD	SMOKE DETECTOR
SPSS	STATIC PRESSURE SENSING STATION
T	TEMPERATURE/ THERMOSTAT
TOT	TOTAL
TSP	TOTAL STATIC PRESSURE
UH	UNIT HEATER
V	VOLTS
VFD	VARIABLE FREQUENCY DRIVE
W	WIDTH / WATTS
WB	WET BULB

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04/15/2021	PERMIT RESPONSE
05/15/2021	BID SUBMISSION
10/15/2021	CONSTRUCTION CONFORMING DOCUMENTS

TITLE:

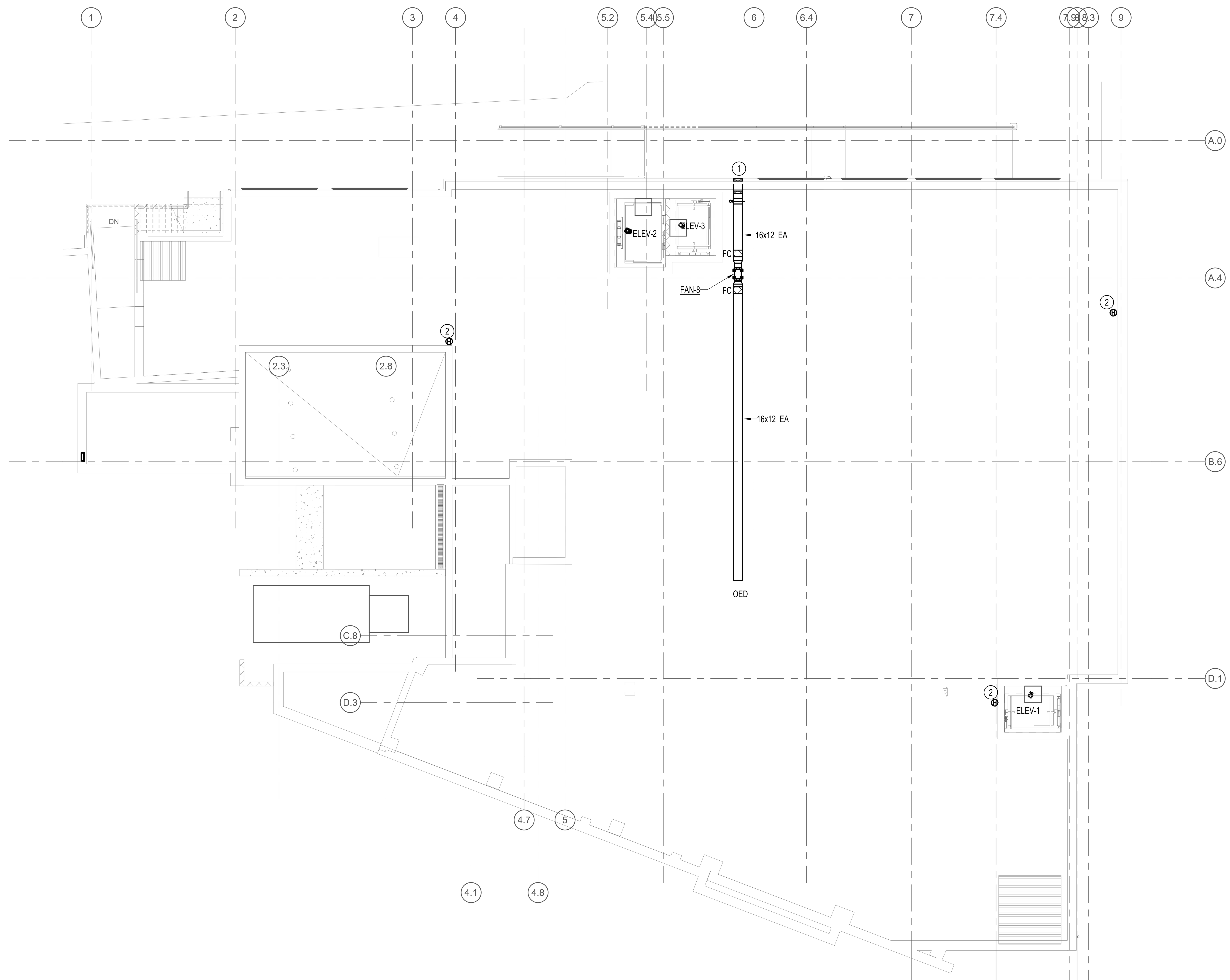
MECHANICAL COVER SHEET

DATE: 10.15.2021

JOB NO.:201907.1

SHEET NO.

M001



1 M100 CRAWL SPACE - MECHANICAL - NEW WORK
1/8" = 1'-0"

DRAWING NOTES: (APPLICABLE TO THIS DRAWING ONLY)

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3. ALL ITEMS THAT REQUIRE ACCESS FOR OPERATING, CLEANING, SERVICING, MAINTENANCE, AND CALIBRATION, SHALL BE EASILY AND SAFELY ACCESSIBLE. EXAMPLES OF THESE ITEMS INCLUDE, BUT ARE NOT LIMITED TO: ALL TYPES OF VALVES, FILTERS, STRAINERS, TRANSMITTERS, AND CONTROL DEVICES. PRIOR TO COMMENCING INSTALLATION WORK, REFER CONFLICTS BETWEEN THIS REQUIREMENT AND CONTRACT DRAWINGS TO THE COR FOR RESOLUTION.

SPECIAL NOTES: (APPLICABLE TO THIS DRAWING ONLY)

1. CONNECT DUCT 16x16 WALL LOUVER (UNDER WALKWAY RAMP) AS SPECIFIED BY ARCHITECT. PROVIDE NECESSARY TRANSITION.
2. SPACE MOUNTED HUMIDISTAT. FAN SHALL ENERGIZE WHEN HUMIDITY IN CRAWLSPACE IS ABOVE 60%.

0 4 8 16 24 FT
1/8" = 1'-0"

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05/15/2021	BID SUBMISSION
10/15/2021	CONSTRUCTION CONFORMING DOCUMENTS

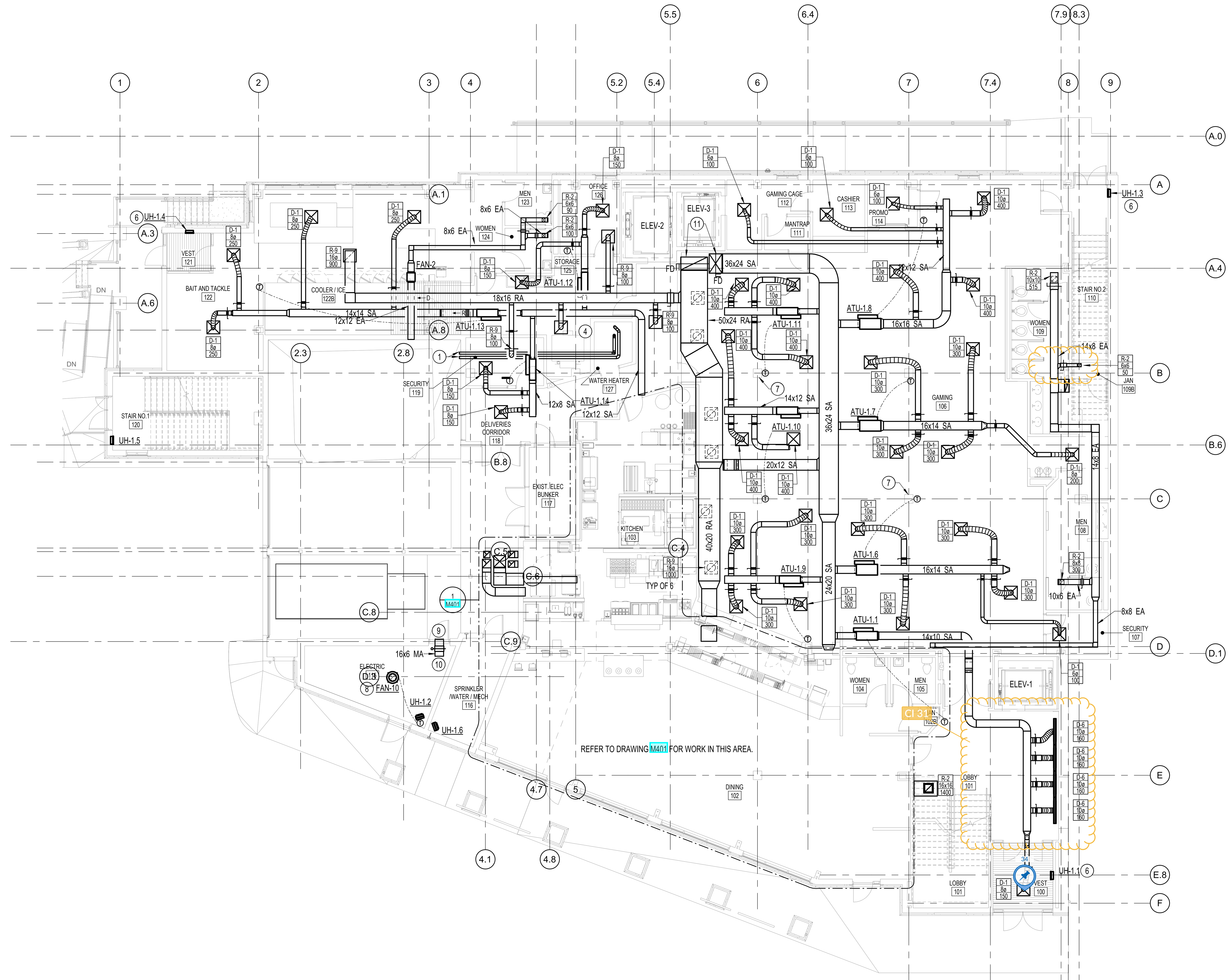
TITLE:

CRAWL SPACE - MECHANICAL - NEW WORK

DATE: 10.15.2021
JOB N0.:201907.1

SHEET NO.

M100



1 M101 FIRST FLOOR PLAN - MECHANICAL - NEW WORK
1/8" = 1'-0"

DRAWING NOTES: (APPLICABLE TO THIS DRAWING ONLY)

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4. IF DIFFUSER RUNOUT IS NOT INDICATED THEN RUNOUT SIZE IS SAME AS DEVICE NECK SIZE.

SPECIAL NOTES: (APPLICABLE TO THIS DRAWING ONLY)

1. TERMINATE WITH CONCENTRIC VENT KIT.
2. 36x24 SA, 50x30 RA DUCTS UP TO 2ND FLR.
3. 24x8 EXHAUST AIR DUCT UP TO FLOOR ABOVE.
4. MOUNT ON HOUSEKEEPING PAD.
5. SUPPORT FROM STRUCTURE ABOVE.
6. MOUNT ±18" AFF TO BOTTOM OF UNIT.
7. MOUNT CARBON DIOXIDE SENSOR HERE FOR OUTSIDE AIR CONTROL. SEE SHEET M701 FOR MORE INFORMATION.
8. MOUNT FAN ON ROOF ABOVE (BELOW STEEL CATWALK PLATFORM). EXTEND 10x10 EXHAUST DUCT DOWN TO ELECTRICAL ROOM AND TERMINATE ±9'-0" ABOVE FINISHED FLOOR, OPEN END WITH BIRDSCREEN.
9. 16x6 WALL LOUVER WITH BIRDSCREEN. PROVIDE PLENUM.
10. OPEN END DUCT WITH BIRDSCREEN.
11. 36x24 SA, 50x24 RA DUCTS UP TO FLOOR ABOVE.

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05/15/2021	BID SUBMISSION
07/08/2021	OSFM COMMENT RESPONSE
10/15/2021	CONSTRUCTION CONFORMING DOCUMENTS

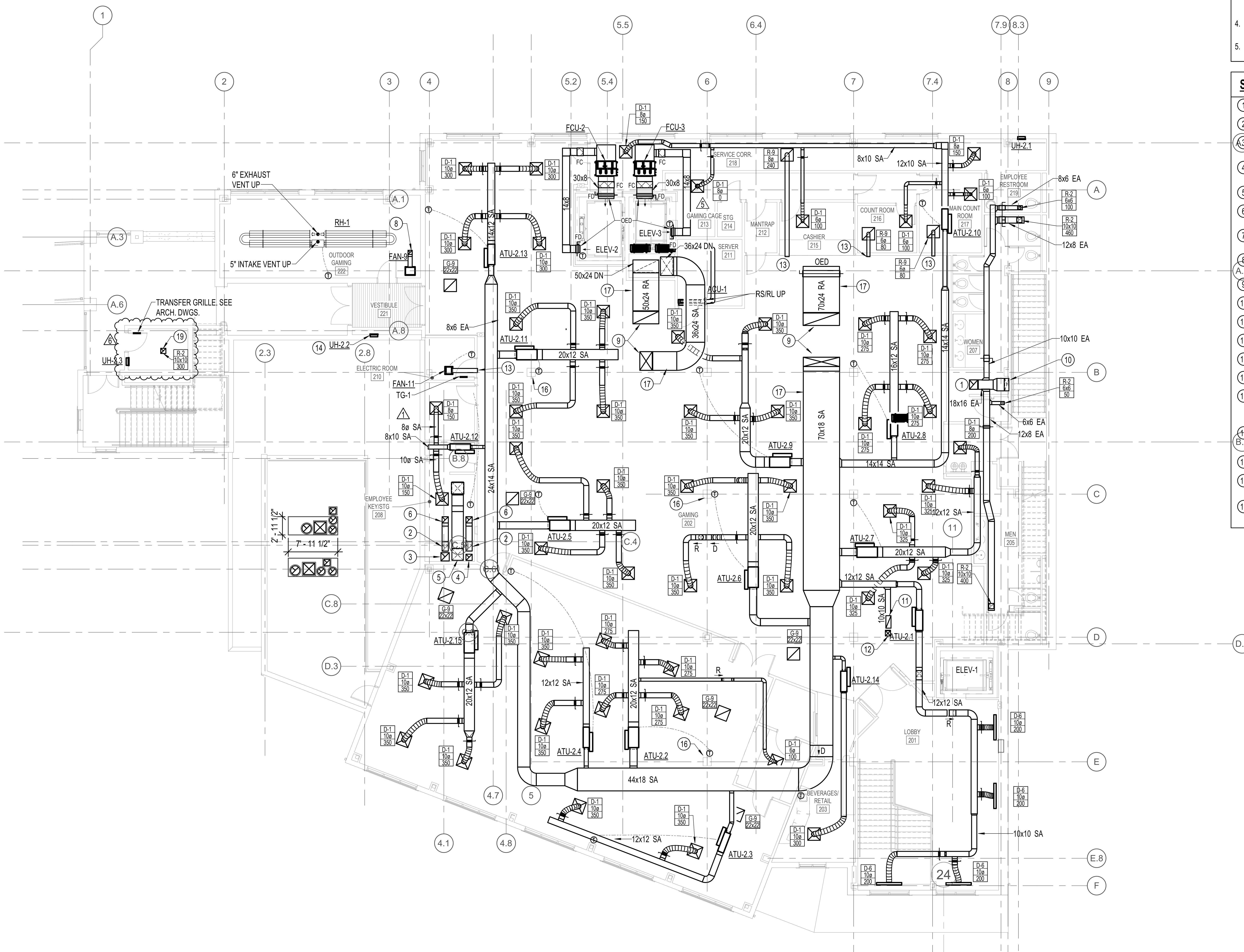
TITLE:

FIRST FLOOR - MECHANICAL - NEW WORK

DATE: 10.15.2021
JOB NO.:201907.1

SHEET NO.

M101



1 M102 SECOND FLOOR PLAN - MECHANICAL - NEW WORK
1/8" = 1'-0"

DRAWING NOTES: (APPLICABLE TO THIS DRAWING ONLY)

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4. IF DIFFUSER RUNOUT IS NOT INDICATED THEN RUNOUT SIZE IS SAME AS DEVICE NECK SIZE.
5. CEILING SPACE IS USED AS RETURN AIR PLENUM. NO PVC PIPING PERMITTED IN PLENUM.

SPECIAL NOTES: (APPLICABLE TO THIS DRAWING ONLY)

1 18x16 EXHAUST AIR UP TO EXHAUST FAN (EF-1).
2 12x12 EXHAUST AIR DOWN IN SHAFT.
3 16x16 EXHAUST AIR DOWN IN SHAFT AND UP TO EXHAUST FAN (KEF-3) ABOVE. PROVIDE NECESSARY TRANISTIONS FOR CONNECTION.
4 12x12 EXHAUST AIR DOWN IN SHAFT AND UP TO EXHAUST FAN (KEF-2) ABOVE. PROVIDE NECESSARY TRANSITIONS FOR CONNECTION.
5 22x22 MAKEUP AIR DOWN IN SHAFT.
6 12x12 EXHAUST DUCT UP TO EXHAUST FAN ABOVE. PROVIDE NECESSARY TRANSITIONS FOR CONNECTION.
7 22x22 MAKEUP DUCT UP TO MUA-1 ABOVE. PROVIDE NECESSARY TRANSITIONS FOR CONNECTION.
8 8x8 EA DUCT UP THRU ROOF. TERMINATE COMPLETE WITH GOOSENECK. FURNISH WITH BIRDSCREEN.
9 SUPPLY AND RETURN AIR DUCT UP TO RTU PLENUM CURB.
10 24x8 EXHAUST AIR DUCT DOWN TO FLOOR BELOW.
11 24x10 OPEN END RETURN DUCT ABOVE CEILING. EXTEND UP TO FLOOR ABOVE.
12 10x10 SUPPLY DUCT UP TO FLOOR ABOVE.
13 OPEN END DUCT.
14 MOUNT ±18" AFF TO BOTTOM OF UNIT.
15 10x10 EXHAUST DUCT UP TO ROOF MOUNTED EXHAUST FAN INLET. PROVIDE NECESSARY TRANSITION. TERMINATE DUCT ±12" FROM TOP OF SHAFT, OPEN ENDED WITH BIRDSCREEN.
16 MOUNT CARBON DIOXIDE SENSOR HERE FOR OUTSIDE AIR CONTROL. SEE SHEET M701 FOR MORE INFORMATION.
17 DUCT SMOKE DETECTOR LOCATION.
18 MOUNT RH-1 ±10'-0" ABOVE FINISHED FLOOR. MOUNT UNIT VIA CHAIN WITH TURNBUCKLE SUSPENDED FROM STRUCTURE ABOVE.
19 10x10 EA DUCT UP TO ROOF MOUNTED EXHAUST FAN.

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04/15/2021	PERMIT RESPONSE
05/15/2021	BID SUBMISSION
06/04/2021	ADDENDUM 1
10/15/2021	CONSTRUCTION CONFORMING DOCUMENTS
02/18/2022	CHANGE BULLETIN 1
07/22/2022	CHANGE BULLETIN 5
09/15/2022	CHANGE BULLETIN 6

TITLE:

SECOND FLOOR - MECHANICAL - NEW WORK

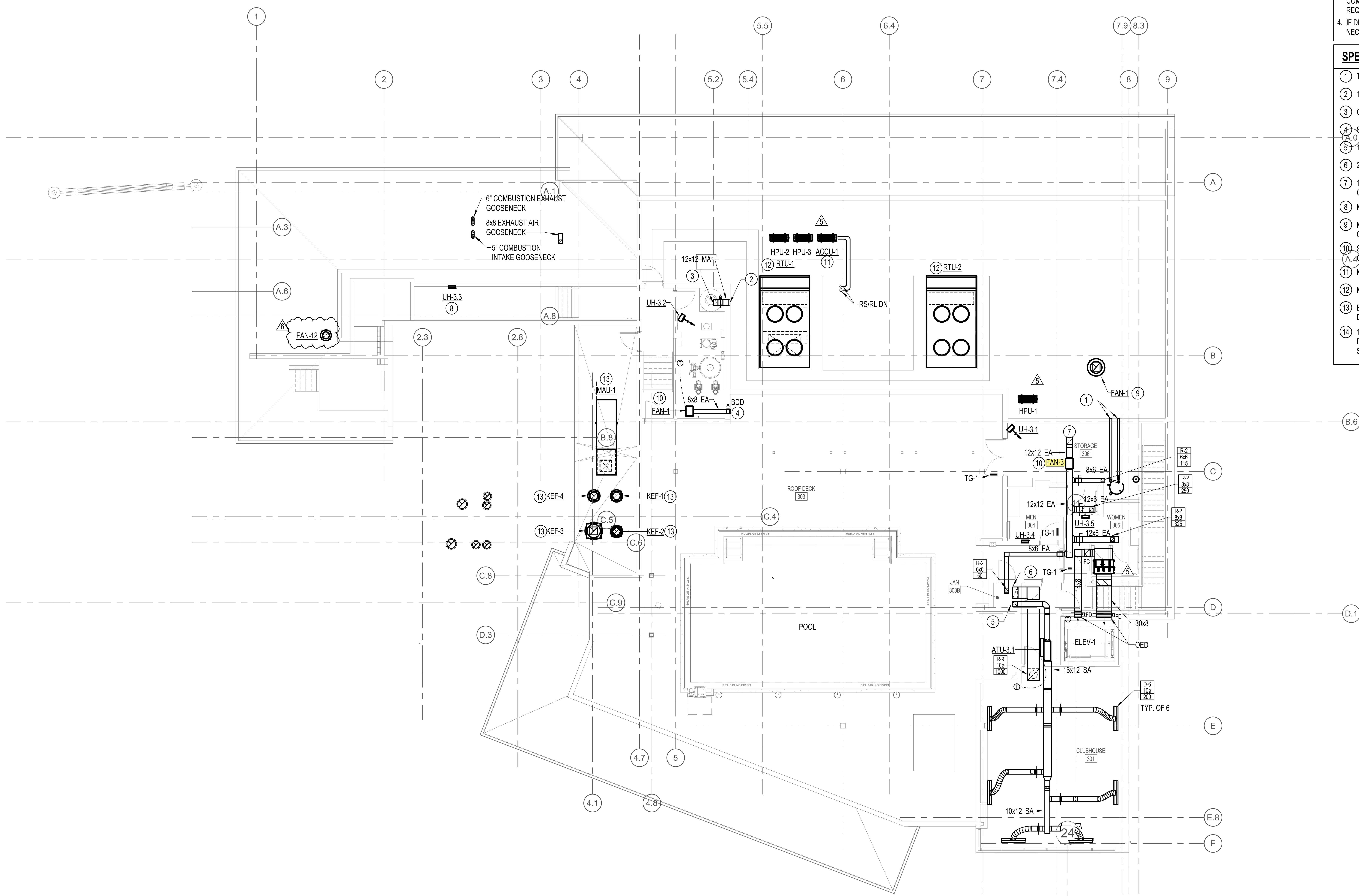
DATE: 10.15.2021

JOB N0.:201907.1

SHEET NO.

M102

1/8" = 1'-0" 0 4 8 16 24 FT



1 ROOF PLAN - MECHANICAL - NEW WORK
M103 1/8" = 1'-0"

DRAWING NOTES: (APPLICABLE TO THIS DRAWING ONLY)

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4. IF DIFFUSER RUNOUT IS NOT INDICATED THEN RUNOUT SIZE IS SAME AS DEVICE NECK SIZE.

SPECIAL NOTES: (APPLICABLE TO THIS DRAWING ONLY)

1 TERMINATE WITH CONCENTRIC VENT KIT.

2 12x12 WALL LOUVER WITH BIRDSCREEN. PROVIDE PLENUM.

3 OPEN END DUCT WITH BIRDSCREEN.

4 8x8 EXHAUST VENT WITH BIRDSCREEN.

5 10x10 SUPPLY DUCT DOWN TO FLOOR BELOW.

6 24x10 RETURN AIR DUCT DOWN TO FLOOR BELOW.

7 12x12 EXHAUST AIR DUCT UP THRU ROOF. TERMINATE COMPLETE WITH GOOSENECK , BIRDSCREEN, & BACKDRAFT DAMPER.

8 MOUNT ±18" AFF TO BOTTOM OF UNIT.

9 MOUNT ON 12" ROOF CURB. CONNECT TO EXHAUST DUCT VIA FLEX CONNECTORS.

10 SUPPORT FROM STRUCTURE ABOVE. CONNECT FAN TO DUCT VIA FLEX CONNECTORS.

11 MOUNT UNIT ON RAILS

12 MOUNT UNIT ON 24" PLENUM CURB.

13 EQUIPMENT FURNISHED UNDER ANOTHER DIVISION AND INSTALLED UNDER DIVISION 23.

14 10x10 EXHAUST DUCT UP TO EXHAUST FAN (FAN-7) ON ROOF ABOVE. CONNECT DUCT TO FAN VIA FLEX CONNECTOR. TERMINATE DUCT ±12" FROM TOP OF SHAFT, OPEN ENDED WITH BIRDSCREEN.

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CERTIFICATION:
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license number 25187, expiration date 01/15/22.

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410.532.2395

CONTRACTORS:
High Construction Company
1853 William Penn Way
Lancaster, PA 17601
717.291.2276

PROJECT:

ROD N' REEL RESORT

4160 MEARS AVE.
CHESAPEAKE BEACH,
MD 20732

REVISIONS:

01/15/2021	FOR PERMIT
04/15/2021	PERMIT RESPONSE
05/15/2021	BID SUBMISSION
10/15/2021	CONSTRUCTION CONFORMING DOCUMENTS
09/15/2022	CHANGE BULLETIN 6

TITLE:

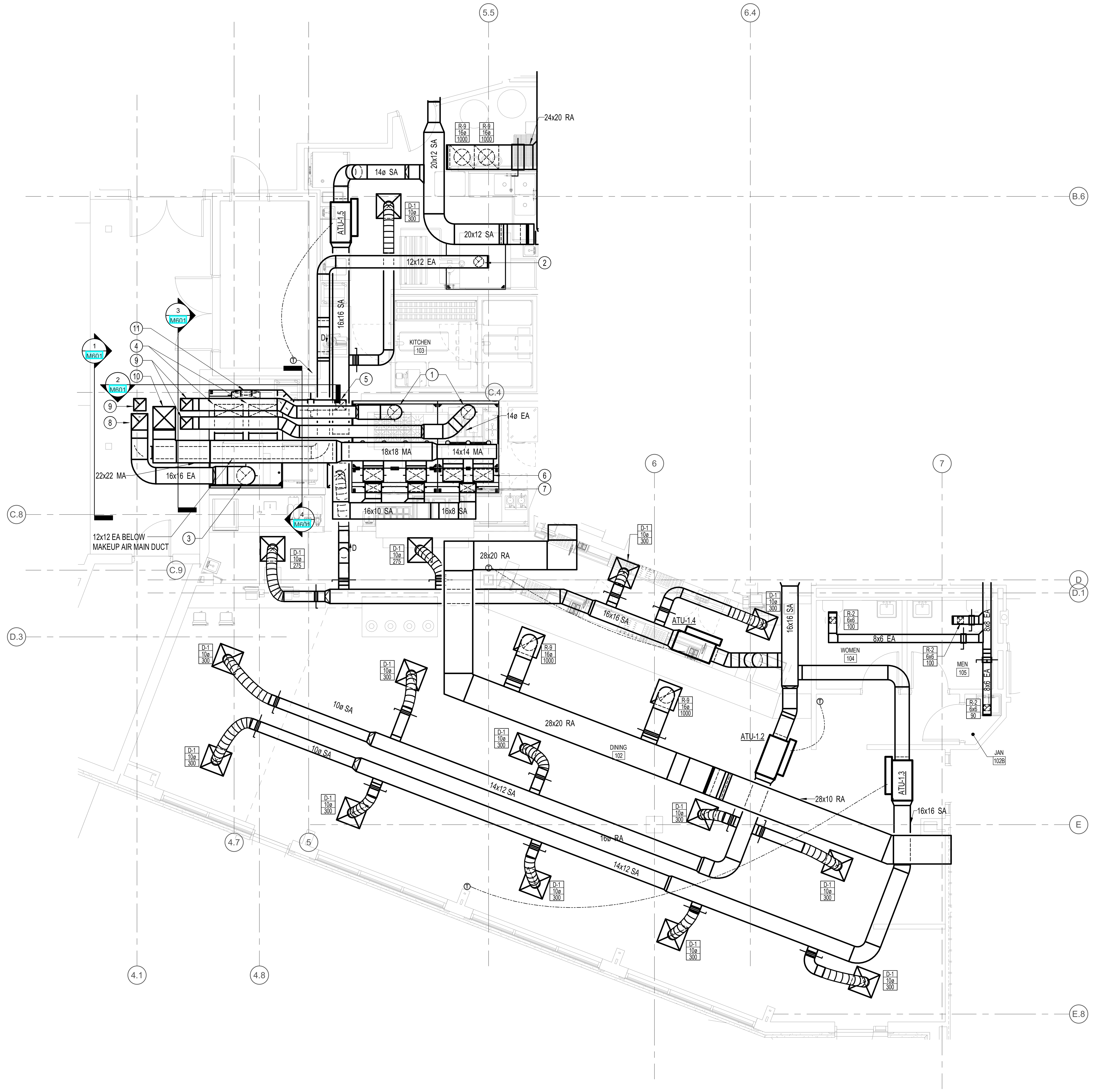
ROOF - MECHANICAL - NEW WORK

DATE: 10.15.2021

JOB N0.:201907.1

SHEET NO.

M103



FIRST FLOOR - CBQ - MECHANICAL HVAC
1/4" = 1'-0"

DRAWING NOTES: (APPLICABLE TO THIS DRAWING ONLY)

1. UNLESS OTHERWISE NOTED, MECHANICAL ITEMS SHOWN HEAVY SOLID (——) SHALL BE NEW, AND LIGHT SOLID (——) ARE EXISTING TO REMAIN.
2. INFORMATION SHOWN ON THIS DRAWING PERTAINING TO EXISTING CONDITIONS HAS BEEN OBTAINED FROM AVAILABLE BUILDING DRAWINGS OR GENERAL FIELD OBSERVATIONS AND MAY NOT INDICATE ACTUAL EXISTING CONDITIONS IN DETAIL OR DIMENSION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR DETERMINING THE ACTUAL EXISTING CONDITIONS PRIOR TO FABRICATION OR PERFORMANCE OF ANY WORK. SHOULD CONDITIONS BE DISCOVERED THAT PREVENT EXECUTION OF THE WORK AS INDICATED, THE CONTRACTOR SHALL IMMEDIATELY NOTIFY THE ENGINEER IN WRITING AND AWAIT WRITTEN DIRECTION BEFORE PROCEEDING WITH THE WORK.
3. ALL ITEMS THAT REQUIRE ACCESS FOR OPERATING, CLEANING, SERVICING, MAINTENANCE, AND CALIBRATION, SHALL BE EASILY AND SAFELY ACCESSIBLE. EXAMPLES OF THESE ITEMS INCLUDE, BUT ARE NOT LIMITED TO: ALL TYPES OF VALVES, FILTERS, STRAINERS, TRANSMITTERS, AND CONTROL DEVICES. PRIOR TO COMMENCING INSTALLATION WORK, REFER CONFLICTS BETWEEN THIS REQUIREMENT AND CONTRACT DRAWINGS TO THE COR FOR RESOLUTION.

SPECIAL NOTES: (APPLICABLE TO THIS DRAWING ONLY)

1. 14Ø EXHAUST DUCT DOWN TO HOOD. MAKE CONNECTION AS REQUIRED BY MANUFACTURER. REFER TO KITCHEN CONSULTANTS DRAWINGS.
2. 10Ø EXHAUST DUCT DOWN TO HOOD. MAKE CONNECTION AS REQUIRED BY MANUFACTURER. REFER TO KITCHEN CONSULTANTS DRAWINGS.
3. 18Ø EXHAUST DUCT DOWN TO HOOD. MAKE CONNECTION AS REQUIRED BY MANUFACTURER. REFER TO KITCHEN CONSULTANTS DRAWINGS.
4. TAP 28X12 MA DUCT AND EXTEND TO HOOD MAKEUP AIR CONNECTIONS. MAKE CONNECTION AS REQUIRED BY MANUFACTURER. REFER TO KITCHEN CONSULTANTS DRAWINGS.
5. TAP 10Ø SA DUCT ABOVE MAIN SA AND EXTEND TO HOOD SUPPLY AIR CONNECTION. PROVIDE NECESSARY TRANSITION. MAKE CONNECTION AS REQUIRED BY MANUFACTURER. REFER TO KITCHEN CONSULTANTS DRAWINGS.
6. 20X12 MA DUCT DOWN TO HOOD MAKEUP AIR CONNECTION. PROVIDE VOLUME DAMPER. MAKE CONNECTION AS REQUIRED BY MANUFACTURER. REFER TO KITCHEN CONSULTANTS DRAWINGS. TYPICAL OF 4.
7. 16X8 SA DUCT DOWN TO HOOD SUPPLY AIR CONNECTION. PROVIDE VOLUME DAMPER. MAKE CONNECTION AS REQUIRED BY MANUFACTURER. REFER TO KITCHEN CONSULTANTS DRAWINGS. TYPICAL OF 3.
8. 16X16 EA DUCT UP IN CHASE TO ROOF.
9. 12X12 EA DUCT UP IN CHASE TO ROOF.
10. 22X22 MA DUCT UP IN CHASE TO ROOF.
11. TRANSITION 10X10 SA DUCT TO 28X6 AND CONNECT TO HOOD SUPPLY AIR CONNECTION.

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05/15/2021	BID SUBMISSION
06/04/2021	ADDENDUM 1
10/15/2021	CONSTRUCTION CONFORMING DOCUMENTS

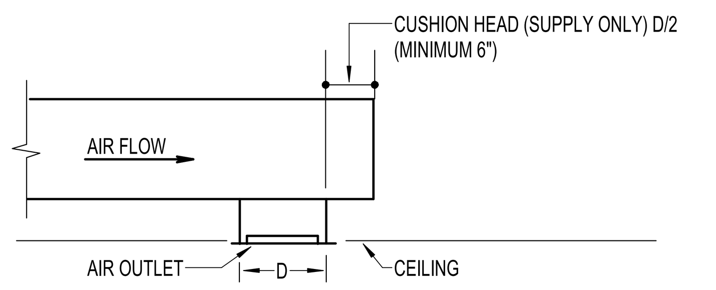
TITLE:

ENLARGED MECHANICAL PLANS

DATE: 10.15.2021
JOB N0.:201907.1

SHEET NO.

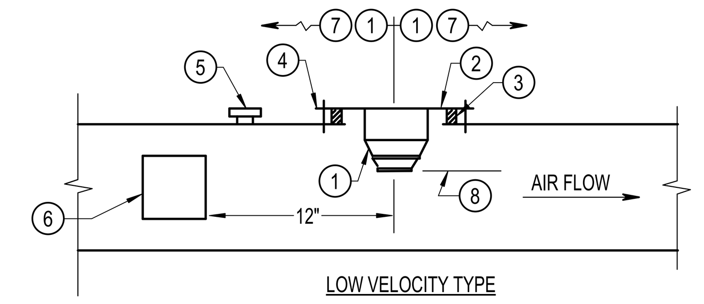
M401



NOTES:

1. AIR FLOW IS REVERSE FOR RETURN AND EXHAUST.

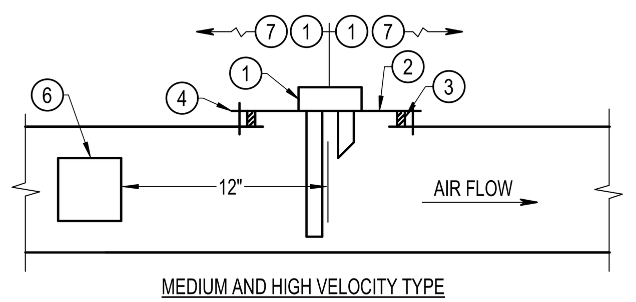
OUTLET LOCATED AT END OF RUN



LOW VELOCITY TYPE

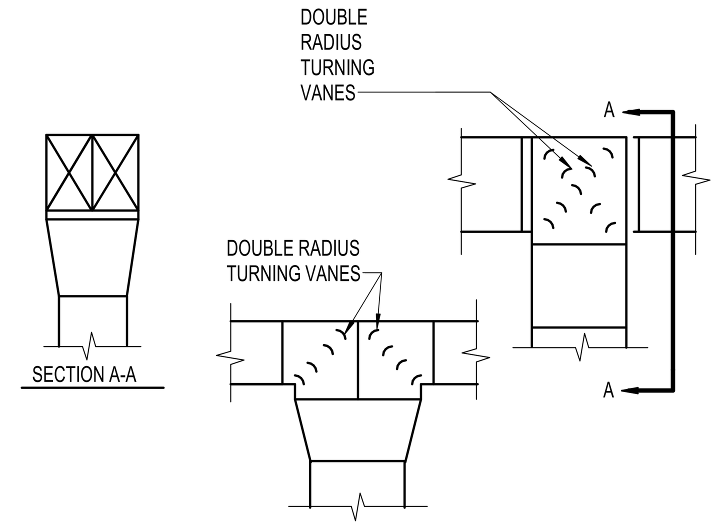
SPECIAL NOTES:

1. SMOKE DETECTOR. INSTALL PER MANUFACTURER'S RECOMMENDATIONS.
2. 10 GAUGE PLATE
3. CONTINUOUS GASKET
4. SHEET METAL SCREWS. TWO PER SIDE.
5. AIR FLOW MEASUREMENT PORT. (INSTRUMENT TEST HOLE)
6. ACCESS DOOR. 6" X 6" IN 14" AND SMALLER DUCTS. 12" X 12" IN LARGER DUCTS.
7. MINIMUM ONE DUCT PERIMETER TO OFFSET OR FITTING
8. MINIMUM 1-1/4" PROJECTION INTO AIRSTREAM. MAXIMUM 1/3 DUCT DEPTH

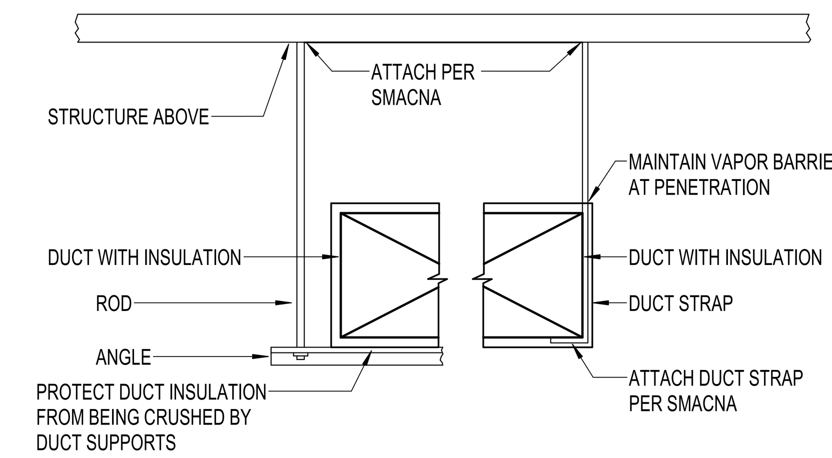


MEDIUM AND HIGH VELOCITY TYPE

SMOKE DETECTOR INSTALLATION RECTANGULAR DUCT



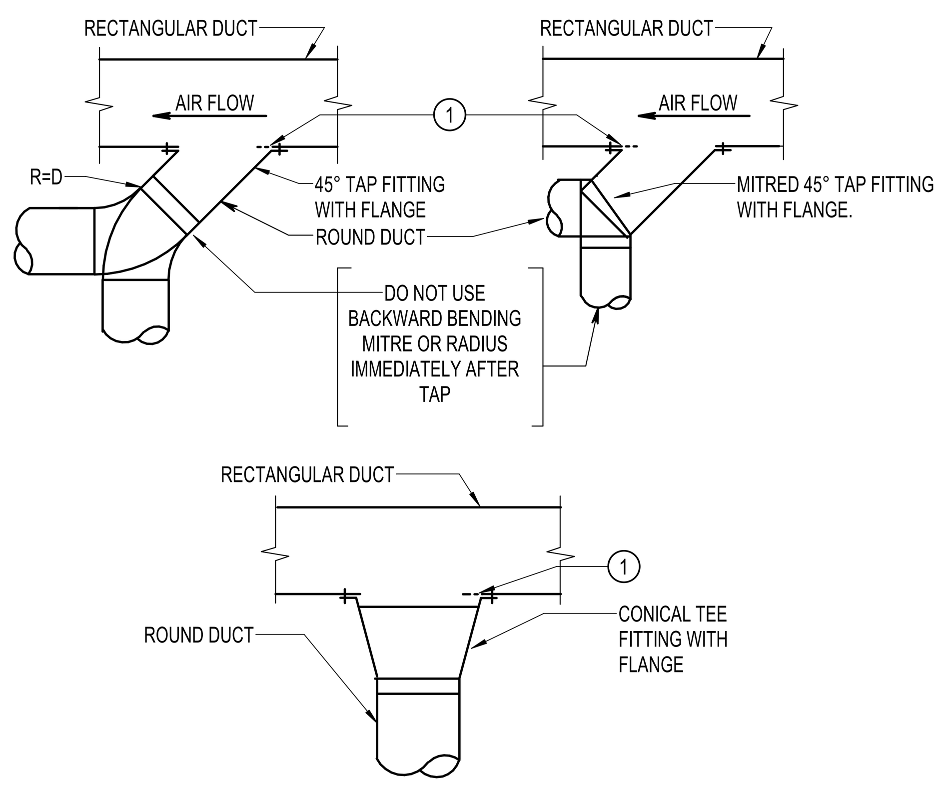
DUCT SPLITS



NOTES:

1. REFER TO SMACNA FOR ADDITIONAL HANGERS AND SUPPORTS

DUCT INSULATION SUPPORT DETAIL



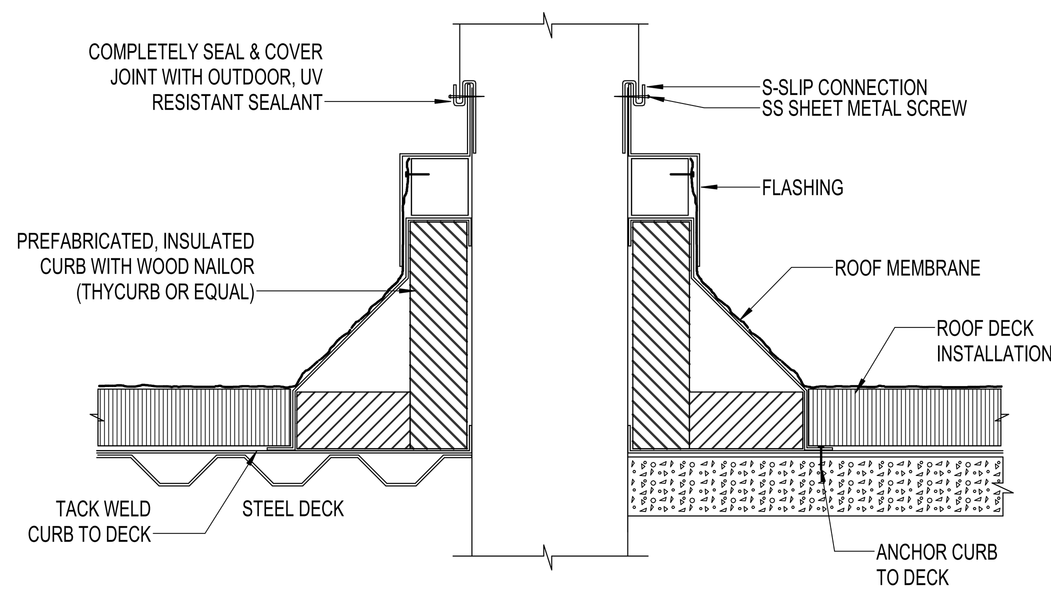
NOTES:

1. DO NOT USE 45° TAP FITTINGS EXCEPT WHERE TIGHT CONDITION PREVENTS USE OF OTHER FITTINGS INDICATED.

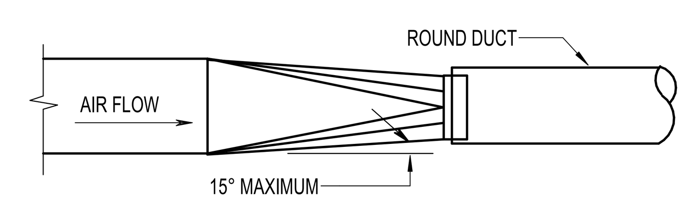
SPECIAL NOTES:

1. CUT HOLE IN RECTANGULAR DUCT LARGE ENOUGH SO THAT ROUGH METAL EDGE DOES NOT PROJECT INTO AIR STREAM.

TAPOFF FROM RECTANGULAR DUCT TO ROUND DUCT



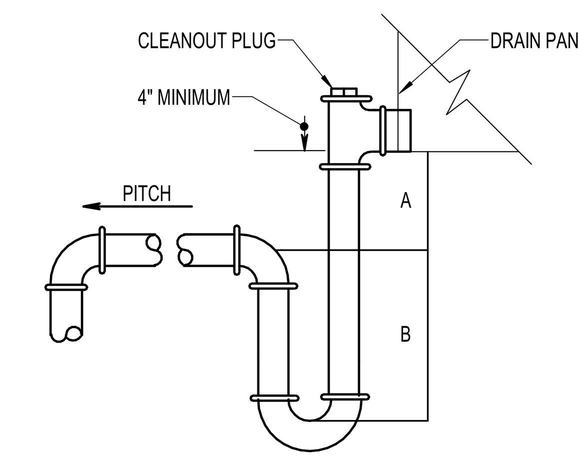
DUCT PENETRATION THROUGH ROOF



NOTES:

1. FOR AIR FLOW IN OPPOSITE DIRECTION, TRANSITION COLLAR SHALL OVERLAP EXTERIOR OR ROUND DUCT.

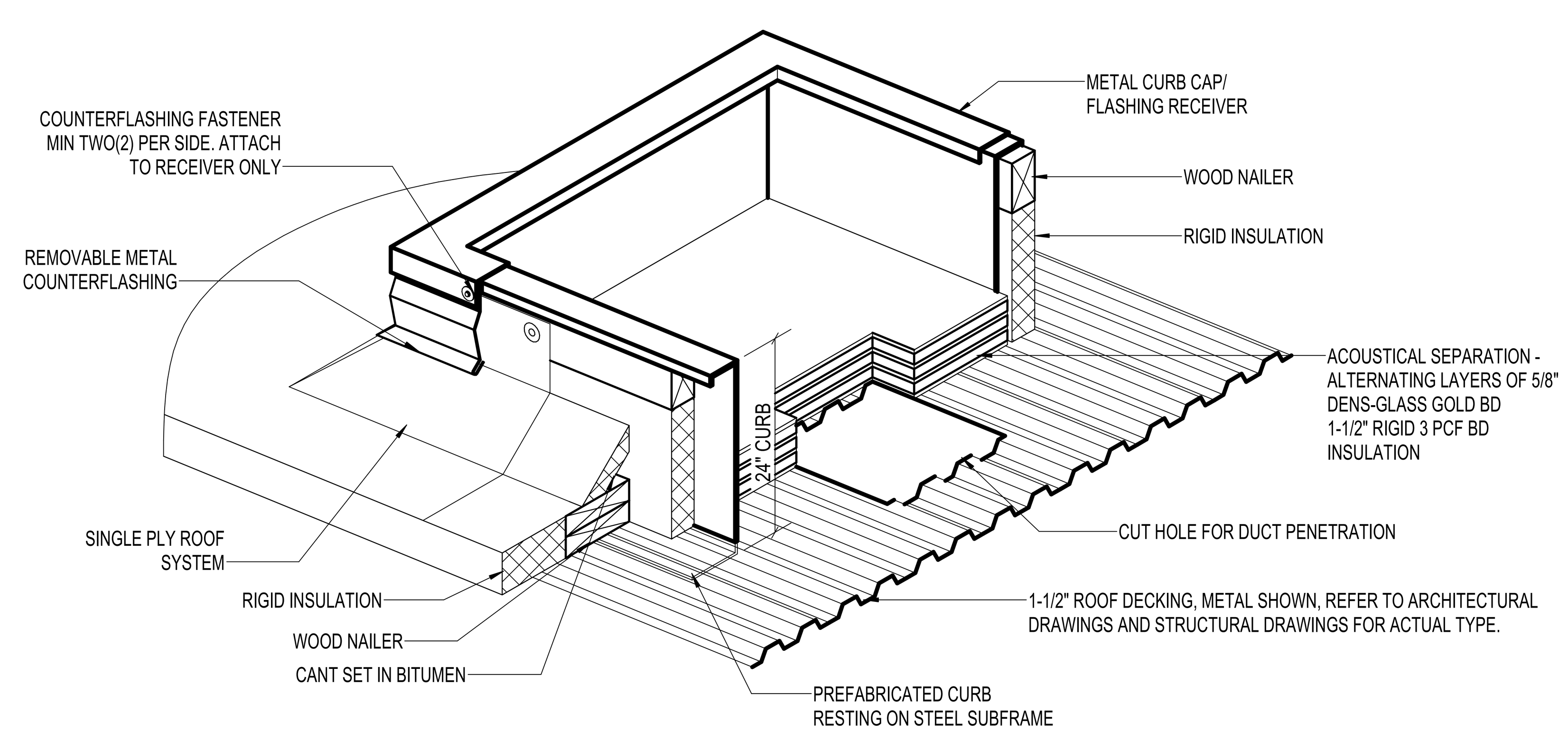
RECTANGULAR TO ROUND DUCT TRANSITION



UNIT TYPE	A	B
DRAW THRU	2" PLUS X	X
BLOW THRU	1" MINIMUM	2X

WHERE X = STATIC PRESSURE IN FAN

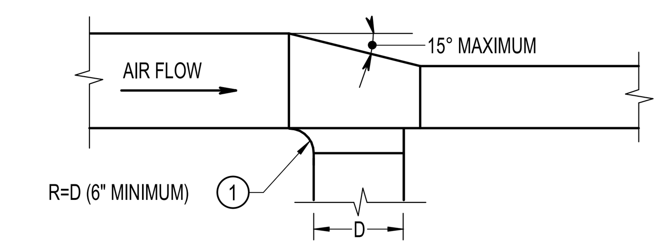
CONDENSATE DRAIN TRAP



NOTES:

1. MAKE ALL LINE, PIPE, DUCT AND CONDUIT PENETRATIONS INSIDE THE CURB. (INCLUDING CONDUIT FOR CONVENIENCE OUTLETS)
2. THIS DETAIL SHOWS THE CURB RESTING DIRECTLY ON THE ROOF DECK. SEE STRUCTURAL FOR ALL SUPPORT REQUIREMENTS.

RTU CURB INSTALLATION DETAIL



NOTES:

1. SEE SMACNA MANUAL FOR METHODS OF SECURING TAP-OFF CONNECTION TO MAIN.
2. SAME FOR RETURN AND EXHAUST DUCTS EXCEPT AIR FLOW IS REVERSED.

SPECIAL NOTES:

1. 45° TAP PERMITTED FOR DUCTWORK RATED 2" W.G. OR LESS.

TAPOFF AND TRANSITION FROM SIDE OF DUCT

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10/15/2021	CONSTRUCTION CONFORMING DOCUMENTS

TITLE:

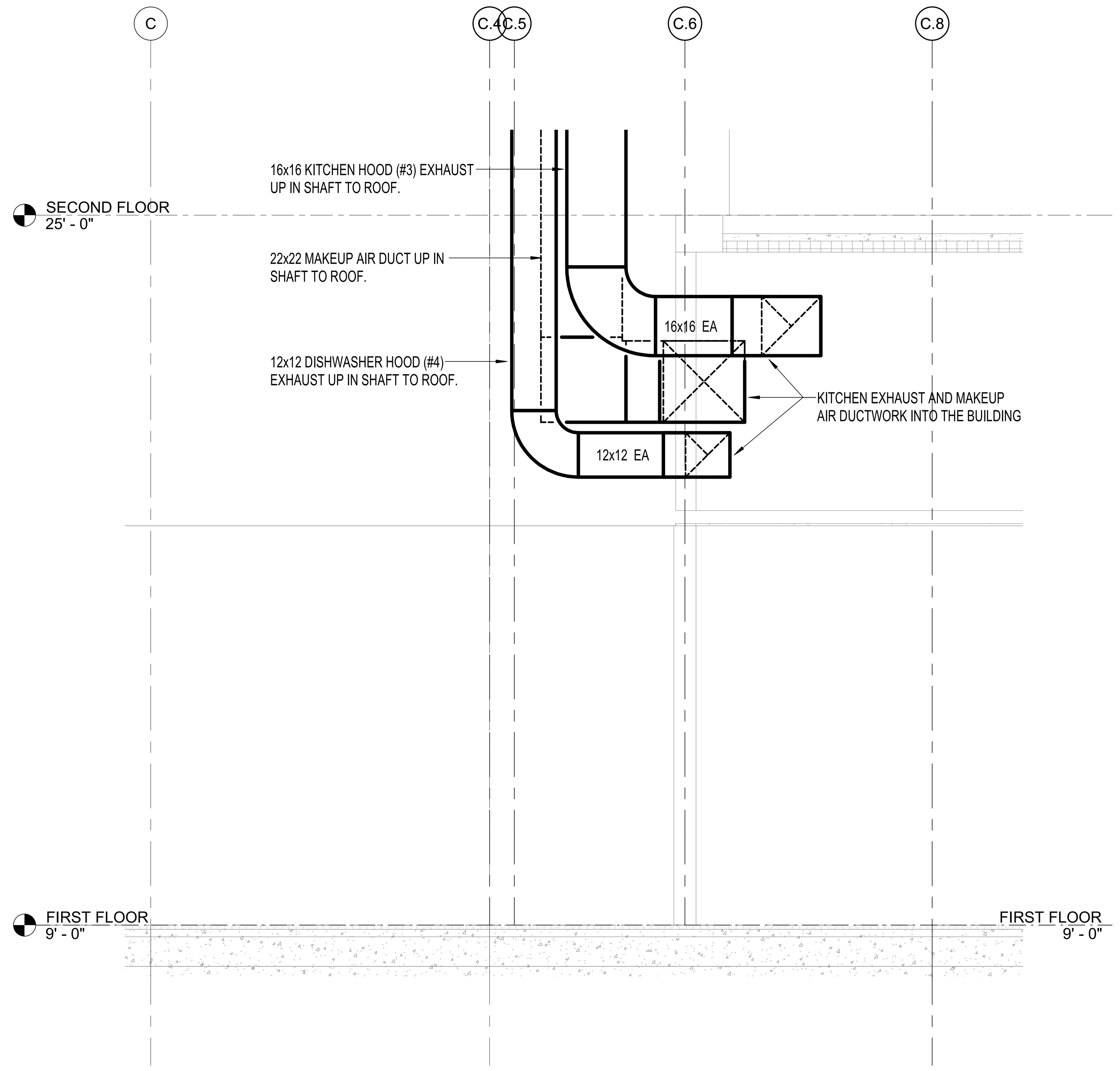
MECHANICAL DETAILS

DATE: 10.15.2021

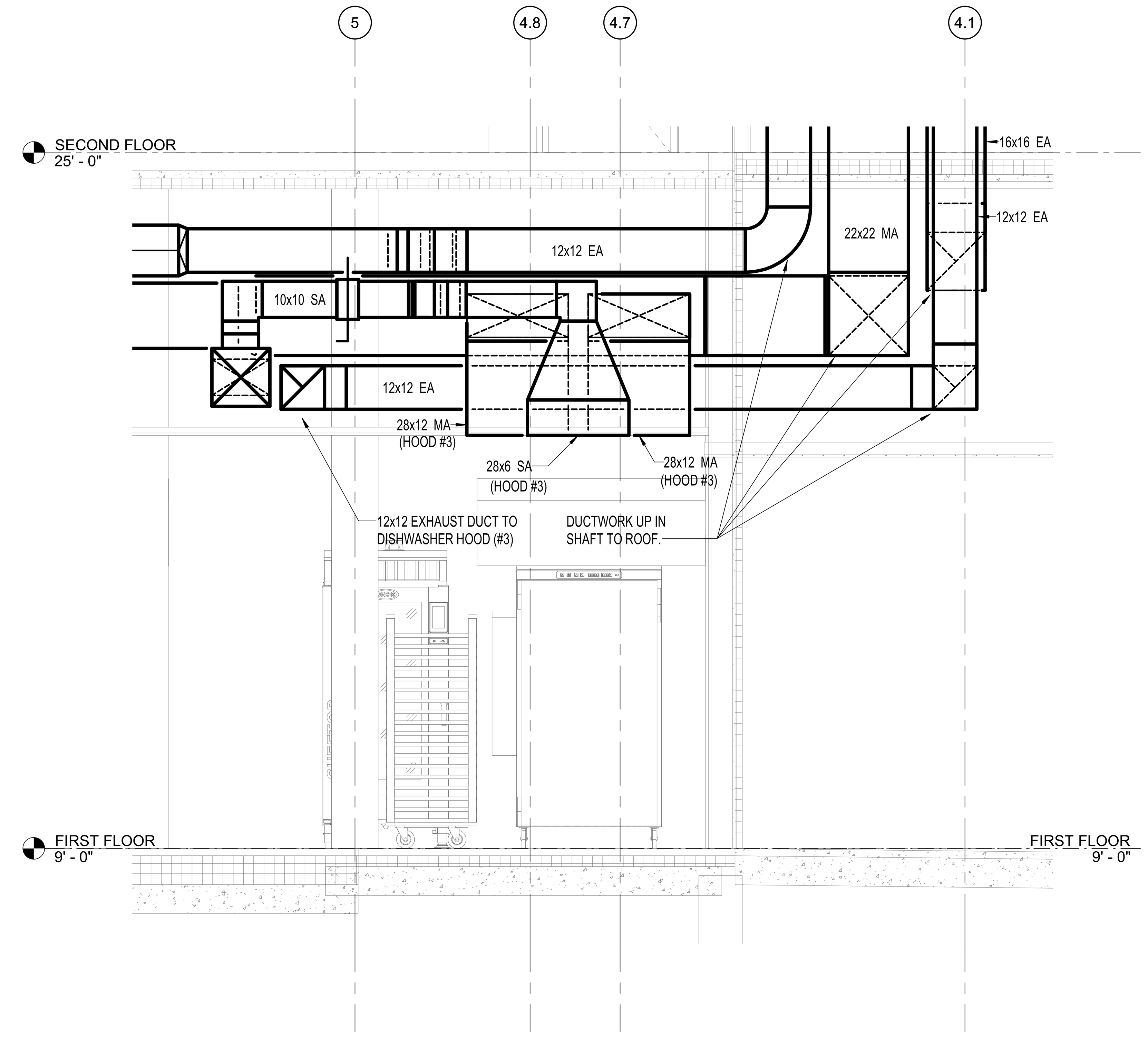
JOB NO.:201907.1

SHEET NO.

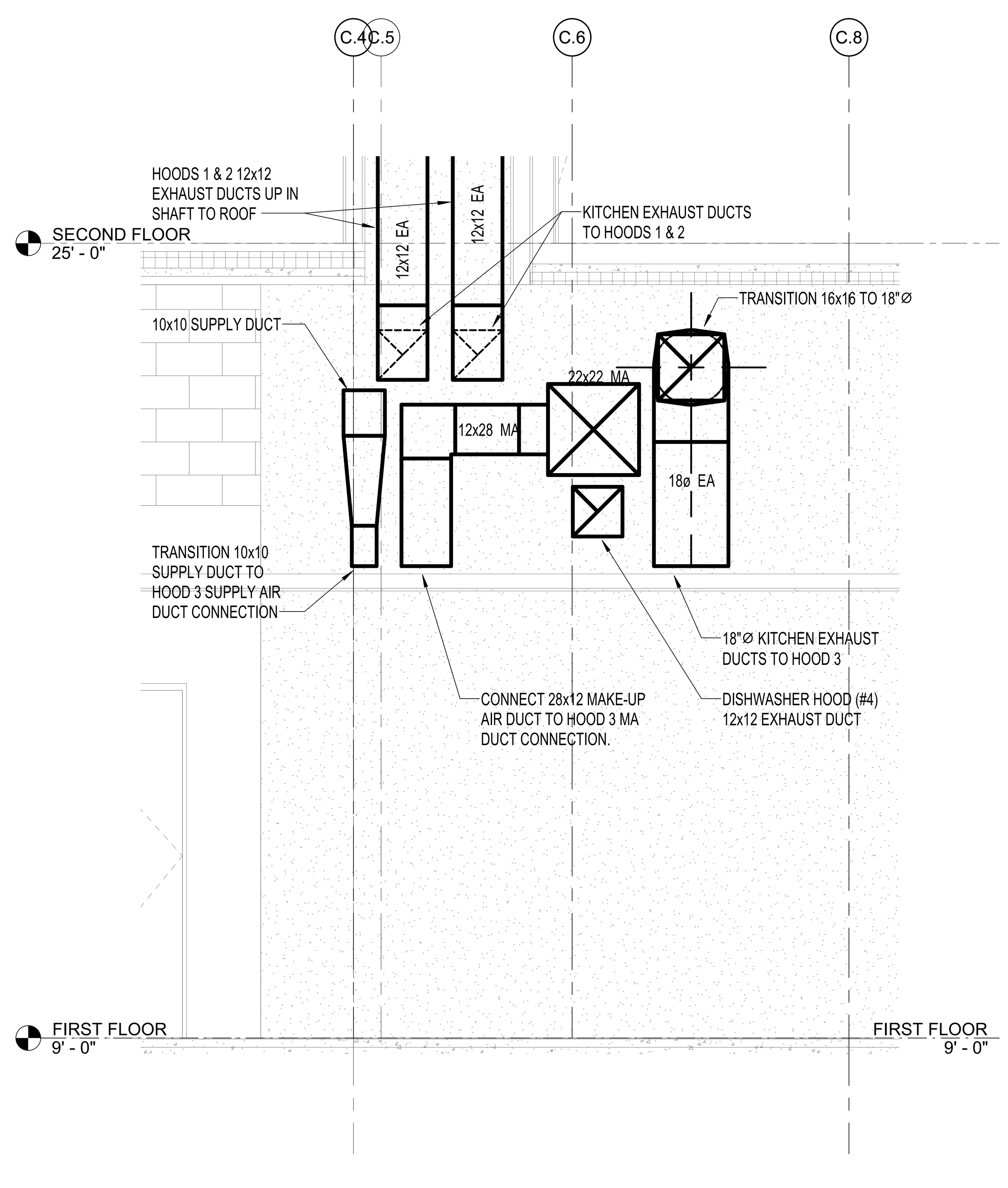
M501



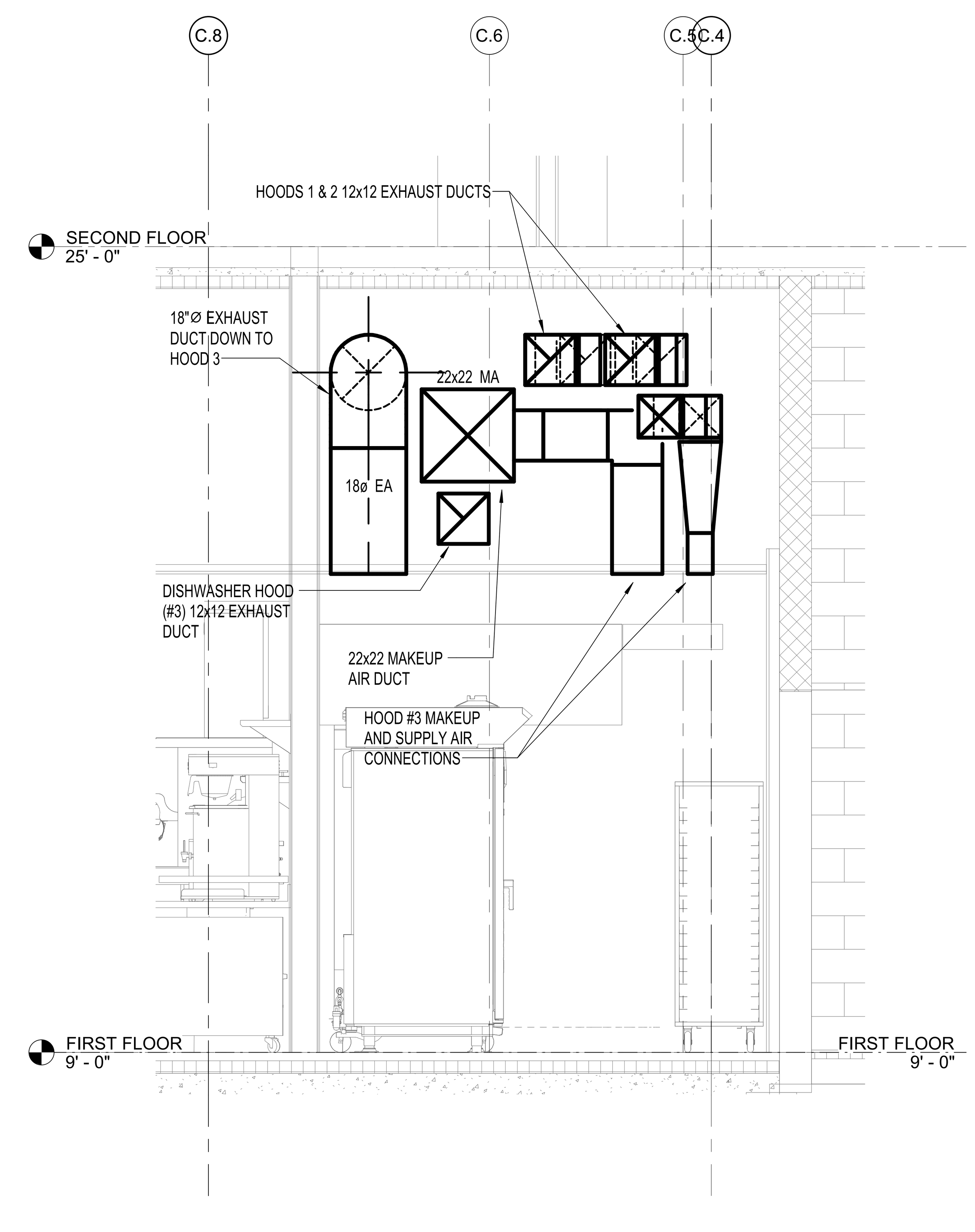
1
M601
KITCHEN EXHAUST DUCTWORKSECTIONS - 1
1/2" = 1'-0"



2
M601
KITCHEN EXHAUST DUCTWORK SECTION - 2
1/2" = 1'-0"



3
M601
KITCHEN EXHAUST DUCTWORK SECTION - 3
1/2" = 1'-0"



4
M601
KITCHEN EXHUAST DUCTWORK SECTION - 4
1/2" = 1'-0"

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TITLE:
MECHANICAL SECTIONS

DATE: 10.15.2021
JOB N0.:201907.1

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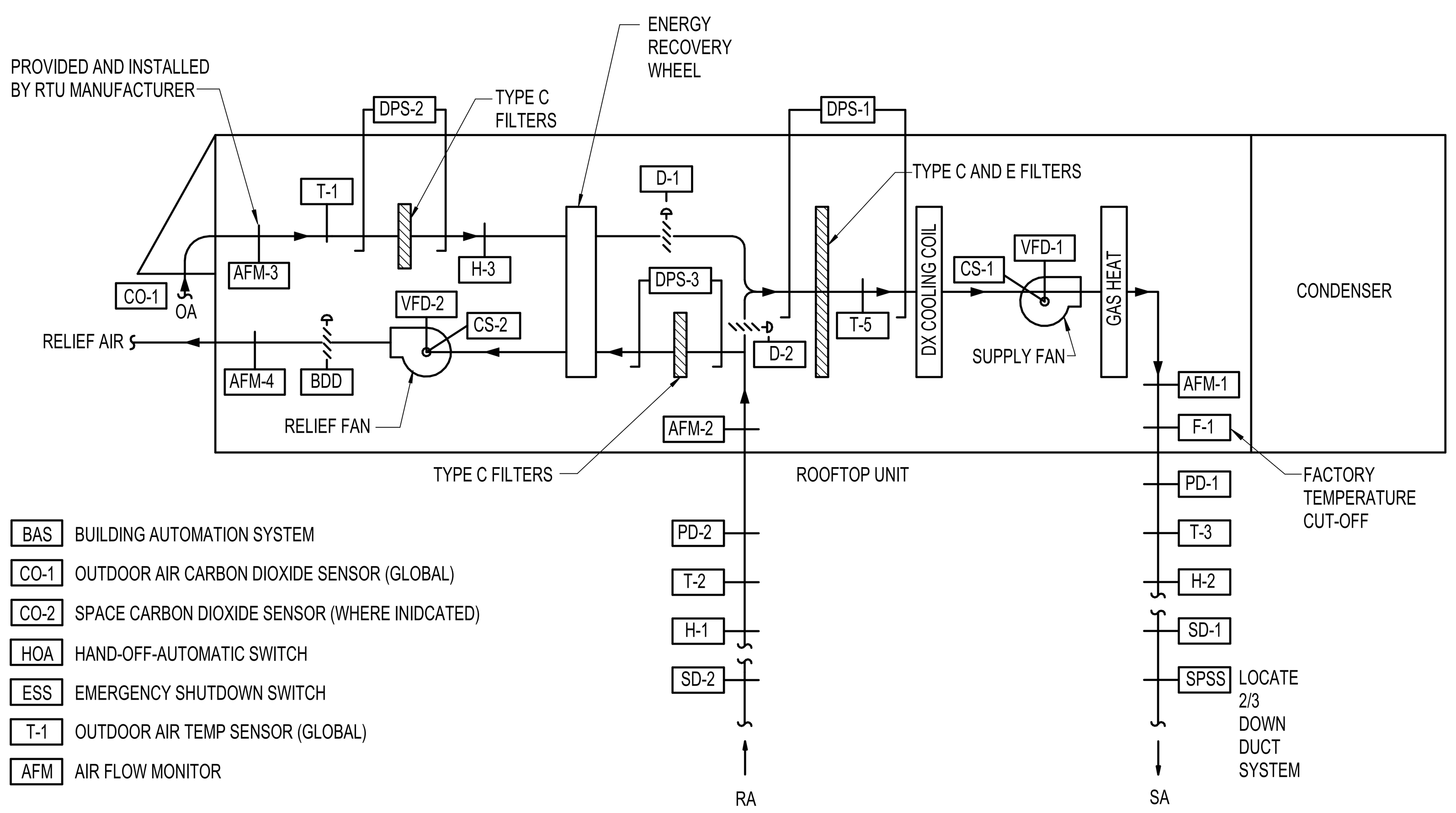
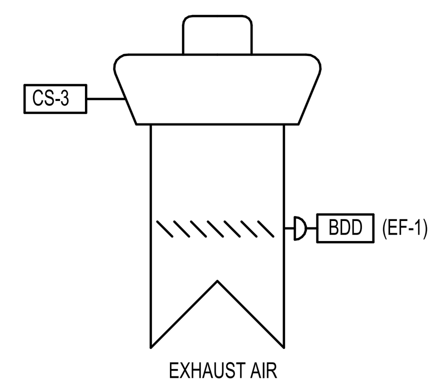
M601

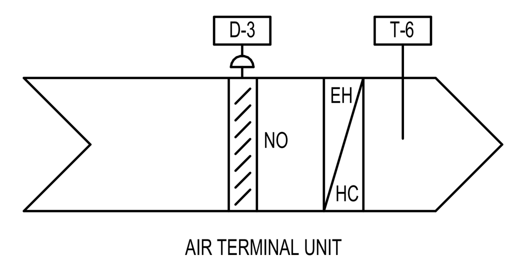


BAS	BUILDING AUTOMATION SYSTEM
CO-1	OUTDOOR AIR CARBON DIOXIDE SENSOR (GLOBAL)
CO-2	SPACE CARBON DIOXIDE SENSOR (WHERE INIDCATED)
HOA	HAND-OFF-AUTOMATIC SWITCH
ESS	EMERGENCY SHUTDOWN SWITCH
T-1	OUTDOOR AIR TEMP SENSOR (GLOBAL)
AFM	AIR FLOW MONITOR



AIRFLOW SCHEDULE

RTU	SUPPLY AIRFLOW		OUTDOOR AIRFLOW		RELIEF AIRFLOW	
	MIN	MAX	MIN	MAX	MIN	MAX
1	4500	13500	1500	5000	0	3500
2	8700	15600	3500	6300	0	3500



CO-2	SPACE CARBON DIOXIDE SENSOR (WHERE INCATED)
BSP-1	BUILDING STATIC PRESSURE SENSOR (WHERE INCATED)
T-4	SPACE TEMPERATURE SENSOR

VARIABLE AIR VOLUME ROOF TOP UNIT CONTROL DIAGRAM

INPUT/OUTPUT SUMMARY

SYSTEM		INDICATION											ALARM				CONTROL					REMARKS
ROOFTOP UINT	GRAPHIC	STATUS	TEMPERATURE	RELATIVE HUNIDITY	PPM	CFM	STATIC PRESSURE	DIFFERENTIAL PRESSURE	GPM	POSITION	RUN TIME	PERCENT	LOW	HIGH	CRITICAL	MAINTENANCE	PROGRAM START/STOP	HAND-OFF-AUTO	MODULATE	MIN/MAX	OPEN/CLOSED	
SYSTEM	●																●	●				BAS, HOA
SUPPLY AIR FAN		●					●				●	●			●		●					CS-1, VFD-1, PD-1
RELIEF AIR FAN		●					●				●	●			●							CS-2, VFD-2
OUTDOOR AIR			●	●		●				●									●			T-1, H-3, AFM-3, D-1
RETURN AIR			●	●		●				●											●	T-2, H-1, AFM-2, D-2
SUPPLY AIR			●	●		●							●		●							T-3, F-1, H-2, AFM-1
RELIEF AIR						●																AFM-4
MIXED AIR			●																			T-5
SMOKE DETECTOR															●							SD-1, SD-2
FILTERS								●						●		●						DPS-1, DPS-2, DPS-3
CARBON DIOXIDE					●																	CO-1, CO-2
SPACE TEMPERATURE			●																			T-4
EMERGENCY SHUTDOWN															●						●	ESS
EXHAUST FAN	●	●													●		●	●				EF-1
ECONOMIZER		●				●				●					●				●		●	RELIEF FAN

VARIABLE AIR VOLUME ROOF TOP UNIT CONTROL DIAGRAM

GENERAL:

1. THE SYSTEM SHALL BE STARTED AND STOPPED THROUGH A SYSTEM HAND-OFF-AUTOMATIC (HOA) SWITCH. WHEN INDEXED TO "HAND", THE SYSTEM SHALL BE ENERGIZED TO OPERATE UNDER THE CONTROL SEQUENCE. WHEN INDEXED TO "OFF", THE SYSTEM SHALL BE DE-ENERGIZED. WHEN INDEXED TO "AUTOMATIC", THE SYSTEM SHALL BE STARTED AND STOPPED THROUGH THE BUILDING AUTOMATION SYSTEM (BAS). THE H-O-A SWITCH POSITION SHALL BE MONITORED THROUGH THE BAS.
2. ALL SAFETIES AND ASSOCIATED CONTROL INTERLOCKS SHALL BE HARDWIRED AND REMAIN ACTIVE IN ALL H-O-A SWITCH POSITIONS.
3. PROVIDE A REMOTE EMERGENCY SHUTDOWN SWITCH (ESS) FOR SYSTEM SHUTDOWN. SWITCH SHALL BE INSTALLED AT A LOCATION APPROVED BY THE FIRE INSPECTOR. TYPICALLY INSTALL AT THE BUILDING ENTRANCE ADJACENT TO THE FIRE ALARM ANNUNCIATOR PANEL.
4. ELECTRIC FREEZESTAT F-1 SHALL SEND UNIT INTO UNOCCUPIED HEATING MODE IF TEMPERATURE FALLS BELOW 45 DEGREES F (ADJUSTABLE). AN ALARM SHALL SOUND THROUGH THE BAS.
5. COOLING SPACE SETPOINT T-4 SHALL BE 75 DEGREES F AND HEATING SPACE SETPOINT T-4 SHALL BE 70 DEGREES F.

STARTING AND STOPPING:

1. THE UNIT IS ENERGIZED TO OPERATE THROUGH THE BAS. CONTROL MANUFACTURER SHALL INTERFACE WITH UNIT TO COMMAND OCCUPIED/UNOCCUPIED/DEHUMIDIFICATION/WARM-UP/COOL-DOWN MODES.
2. PRESSURE SWITCH PD-1 AND/OR PD-2 STOPS SYSTEM SHOULD SUPPLY/RELIEF AIR DUCT PRESSURE RISE ABOVE +/-3" WG (ADJUSTABLE).
3. IF FAN FAILS AS SENSED BY CURRENT SENSOR (CS-1, CS-2), RTU SHALL STOP AND ALARM SHALL SOUND THROUGH BAS.

OCCUPIED (24 HRS) (ADJUSTABLE):

1. RTU IS ENERGIZED. EXHAUST FANS START.
2. RTU MANUFACTURER CONTROLS SHALL MAINTAIN A LEAVING AIR TEMPERATURE OF 55 DEGREES F. MANUFACTURER PACKAGED CONTROLS SHALL INCLUDE DUAL (COMPARATIVE) ENTHALPY BASED ECONOMIZER CONTROL, DX COOLING COIL CONTROL, ENERGY WHEEL CONTROL, AND AUXILIARY HEATING CONTROL. WHEEL BYPASS DAMPERS SHALL BE OPEN TO BYPASS WHEEL DURING ECONOMIZER CONTROL.
3. WHEN FAN IS AT MINIMUM AIRFLOW FOR 30 MINUTES, RESET SUPPLY AIR TEMPERATURE UP 1 DEGREE F. WHEN FAN IS ABOVE MINIMUM AIR FLOW FOR 30 MINUTES, RESET SUPPLY AIR TEMPERATURE DOWN 1 DEGREE F. WHEN RETURN AIR RELATIVE HUMIDITY (H-1) RISES ABOVE 60%, SUPPLY AIR TEMPERATURE SHALL RESET TO 55 DEGREES F UNTIL RETURN RELATIVE HUMIDITY FANS BELOW 50%.

FAN CAPACITY CONTROL:

1. STATIC PRESSURE SENSING STATION SPSS SHALL MODULATE THE SUPPLY AIR FAN SPEED THROUGH THE MANUFACTURER SUPPLIED VARIABLE SPEED CONTROLLER.
2. THE BAS SHALL POLE THE POSITION OF ALL D-3 DAMPERS AND RESET SPSS SETPOINT UNTIL ONE D-3 DAMPER IS OPEN TO 90 PERCENT OF FULL AIR FLOW.

SMOKE DETECTORS:

1. WHEN SMOKE IS SENSED, BY EITHER SMOKE DETECTOR SD-1, OR SD-2 SYSTEM STOPS AND AN ALARM SHALL BE SENT THROUGH BAS. RTU SYSTEM SHALL RETURN TO NORMAL OPERATION WHEN SD-1, AND SD-2 ARE RESET.

OCCUPIED MODE OUTDOOR AIR CONTROL:

1. OUTDOOR AIR SETPOINT SHALL START AT MINIMUM OUTDOOR AIRFLOW IF THE CARBON DIOXIDE DIFFERENTIAL BETWEEN ANY SPACE SENSOR CO-2 AND THE OUTDOOR AIR SENSOR CO-1 EXCEEDS 700 PPM (ADJUSTABLE) THEN VAV(S) SERVING THE SAME SPACE AS THE SPACE SENSOR CO-2 SHALL FULLY OPEN. IF CARBON DIOXIDE DIFFERENTIAL IS STILL ABOVE SETPOINT, RTU IS SET TO MAXIMUM OUTDOOR AIRFLOW (AS SCHEDULED) UNTIL DIFFERENTIAL FALLS BELOW 400 PPM. RTU MINIMUM OUTDOOR AIR SETPOINT IS AS SCHEDULED.

ECONOMIZER:

MANUFACTURER TO PROVIDE THE FOLLOWING INPUTS/OUTPUTS TO CONFORM TO IECC ECONOMIZER CONTROL REQUIREMENTS. WHAT THE RTU MANUFACTURER CANNOT PROVIDE, BAS SHALL INTERFACE AND PROVIDE: DESIRED ECONOMIZER POSITION, ECONOMIZER FEEDBACK POSITION, ECONOMIZER ENABLE SET-POINT, MAXIMUM ECONOMIZER IN HEATING MODE, MINIMUM ECONOMIZER POSITION, MAXIMUM ECONOMIZER CARBON DIOXIDE RESET LIMIT, ECONOMIZER ENABLE STATUS, ECONOMIZER MISSING ALARM, ECONOMIZER FAILURE ALARM. UNIT SHALL BE IN ECONOMIZER MODE, WITH COMPRESSORS OFF, AS LONG AS SPACE TEMPERATURES CAN BE SATISFIED AND HUMIDITY AT H-1 IS BELOW 60% RH.

BUILDING PRESSURIZATION:

BUILDING STATIC PRESSURE IS MEASURED BY THE BUILDING STATIC PRESSURE SENSOR (BSP-1). BUILDING PRESSURE IS MAINTAINED AT A POSITIVE PRESSURE. THE BAS MODULATES RELIEF FAN TO MAINTAIN A BUILDING STATIC PRESSURE OF 0.05" W.G(MAX, ADJ.). WHEN RELIEF FAN IS AT MINIMUM AND BUILDING STATIC PRESSURE IS BELOW 0.02" W.G., BAS SHALL MODULATE OUTDOOR AIR TO MAINTAIN BUILDING PRESSURE SETPOINT.

AIR TERMINAL UNIT CONTROL:

1. DAMPER D-3 SHALL MODULATE FROM MAXIMUM COOLING AIRFLOW TO MINIMUM COOLING AIRFLOW THROUGH VOLUME REGULATOR VR TO MAINTAIN SPACE TEMPERATURE AT T-4 OF 75 DEGREES F. IF A ROOM HAS AN OCCUPANCY SENSOR THAT IS INDEXED OFF, T-4 SHALL BE RESET TO 78 DEGREES F.

2. WHEN SPACE TEMPERATURE AT T-4 FALLS BELOW 70 DEGREES F (ADJUSTABLE), D-3 SHALL BE AT MINIMUM AIR FLOW AND ELECTRIC HEATER SHALL MODULATE TO MAINTAIN SPACE TEMPERATURE. IF A ROOM HAS AN OCCUPANCY SENSOR THAT IS INDEXED OFF, T-4 SHALL BE SET TO 65 DEGREES F. IF A CARBON DIOXIDE SENSOR ALARM, BOX SHALL GO TO MAX COOLING CFM.

3. OUTDOOR AIR CONTROL OVERRIDES AIR TERMINAL UNIT CONTROL. IF VAV(S) ARE FULLY OPEN TO REDUCE CARBON DIOXIDE DIFFERENTIAL BETWEEN SPACE SNESOR C0-2 AND THE OUTDOOR AIR SENSOR CO-1, VAV ELECTRIC HEATER SHALL MODULATE TO MAINTAIN SPACE TEMPERATURE.

VARIABLE AIR VOLUME ROOF TOP UNIT CONTROL SEQUENCE

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DATE: 10.15.2021

JOB N0.:201907.1

SHEET NO.

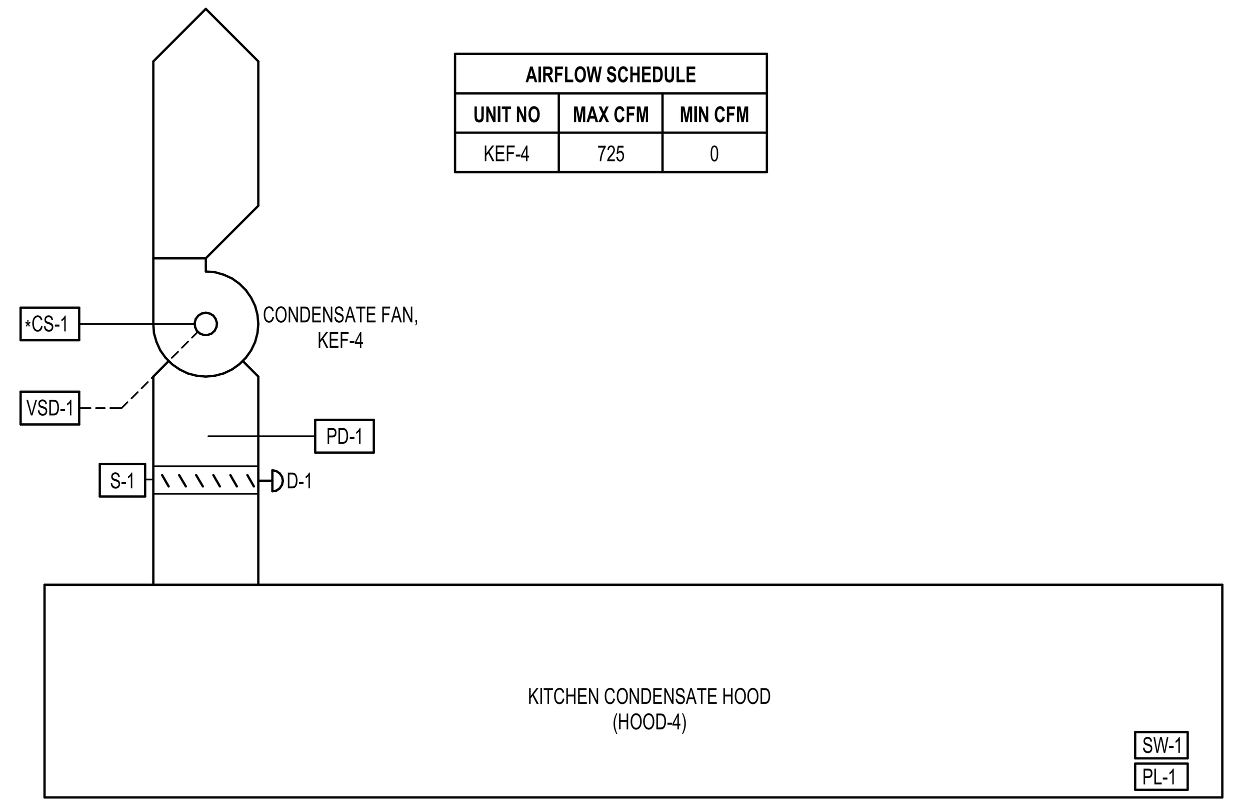
M701

BAS BUILDING AUTOMATION SYSTEM

HOA HAND-OFF-AUTOMATIC

NOTE:
1. D-1 IS PROVIDED BY HOOD MANUFACTURER.
2. DEVICES NOTED WITH ASTERISK PROVIDE READ-OUT INFORMATION ONLY; NO CONTROL FUNCTION

AIRFLOW SCHEDULE		
UNIT NO	MAX CFM	MIN CFM
KEF-4	725	0



KITCHEN CONDENSATE HOOD AND EXHAUST FAN

1. WHEN KITCHEN CONSENSATE HOOD SWITCH, SW-1 IS INDEXED TO ON, D-1 SHALL OPEN. WHEN D-1 IS OPEN, END SWITCH S-1 SHALL START. FAN PILOT LIGHT PL-1 SHALL ILLUMINATE. LOCK-OUT RELAY, ACTIVATED BY FAN STARTER PREVENTS S-1 FROM STOPPING FAN/

2. CONDENSATE EXHAUST FAN SHALL OPERATE THROUGH KITCHEN HOOD MANUFACTURER'S SUPPLIED CONTROLS TO MAINTAIN REQUIRED AIRFLOW.

3. WHEN KITCHEN HOOD IS DE-ENERGIZED, CONDENSATE EXHAUST FAN SHALL DE-ENERGIZE, AND D-1 SHALL CLOSE AFTER 2-MINUTE DELAY (ADJ).

4. PRESSURE DIFFERENTIAL AT SWITCH PD-1 STOPS CONDENSATE EXHAUST FAN IF PRESSURE IN DUCT FALLS BELOW MINUS 2" W.G.

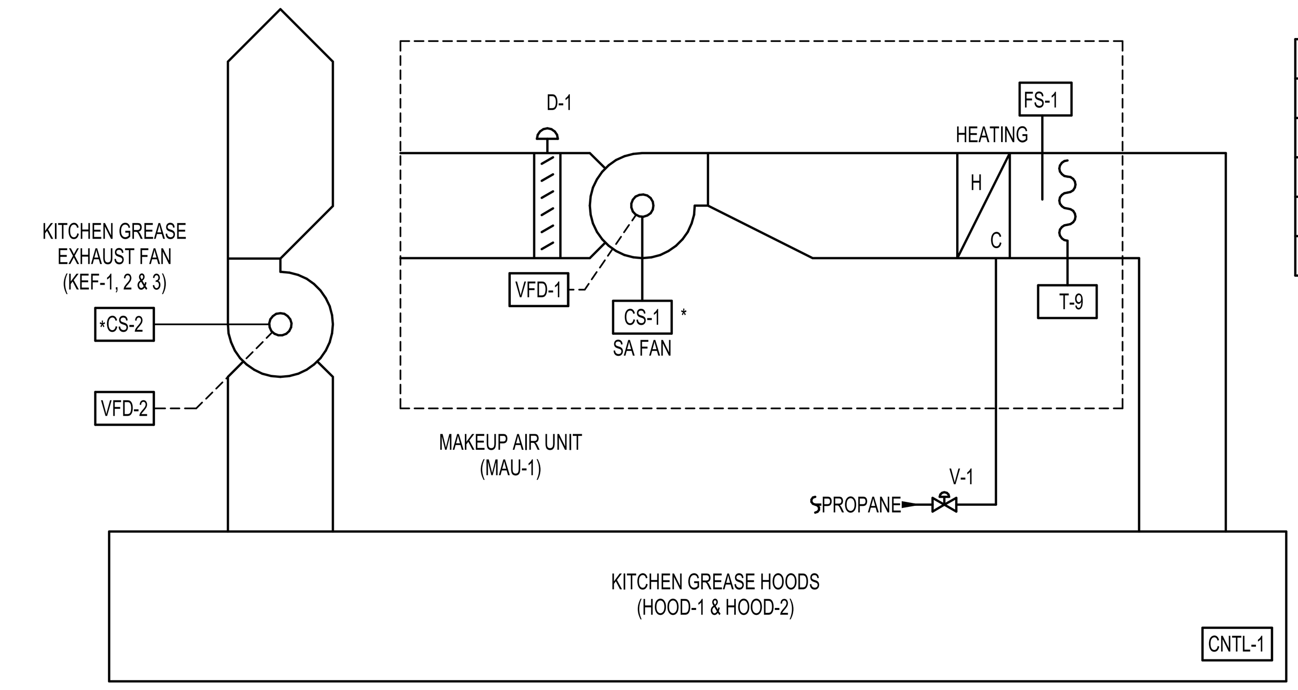
INPUT / OUTPUT SUMMARY

SYSTEM		INDICATION											ALARM				CONTROL					REMARKS
KITCHEN CONDENSATE HOOD AND EXHAUST FAN	GRAPHIC	STATUS	TEMPERATURE	RELATIVE HUMIDITY	PPM	CFM	STATIC PRESSURE	DIFFERENTIAL PRESSURE	GPM	POSITION	RUN TIME	HERTZ	LOW	HIGH	CRITICAL	MAINTENANCE	PROGRAM START /STOP	HAND-OFF-AUTO	MODULATE	MIN/MAX	OPEN/CLOSED	
SYSTEM	●	●																●				BAS, HOA
CONDENSATE FAN		●									●	●	●			●			●			VSD-1, CS-1*, PD-1
CONDENSATE HOOD		●																				SW-1, PL-1
DAMPER										●											●	D-1, S-1

BAS BUILDING AUTOMATION SYSTEM

HOA HAND-OFF-AUTOMATIC

NOTES:
1. DAMPER D-1 IS PROVIDED BY MAU MANUFACTURER.
2. DEVICES NOTED WITH ASTERISK PROVIDE READ-OUT INFORMATION ONLY; NO CONTROL FUNCTION



AIRFLOW SCHEDULE		
UNIT NO	MAX CFM	MIN CFM
MAU-1	3,816	2,000
KEF-1	1,645	862
KEF-2	1,175	611
KEF-3	2,400	1248

KITCHEN HOOD MAKEUP AIR UNIT AND GREASE EXHAUST FAN

1. DEMAND CONTROL VENTILATION PACKAGE (INCL: SMART CONTROLLER, LCD SCREEN NTERFACE, DUCT TEMPERATURE SENSORS, SPACE TEMPERATURE SENSORS, VFDs) SHALL BE PROVIDED BY HOOD MANUFACTURER. INTERLOCK HOOD MANUFACTURER CONTROLS WITH BAS.

2. HOOD SMART CONTROLLER , CNTL-1 SHALL CONSTANTLY MONITOR EACH HOOD EXHAUST AIR TEMPERATURE THROUGH THE RISER MOUNTED TEMPERATURE SENSOR AND CALCULATE DIFFERENTIAL TEMPERATURE COMPARED TO SPACE TEMPERATURE SENSOR.

3. WHEN HOOD TEMPERATURE SENSOR IS 10 DEG. F ABOVE SPACE SENSOR, CONTROLLER SHALL MODULATE EXHAUST FAN SPEED AS SCHEDULED. FAN SHALL MODULATE FROM MIN. TO MAX LINEARLY WITH MIN. AND MAX (25 DEG. F, ADJ.) DIFFERENTIAL TEMPERATURE SETPOINT.

4. EXCEPT IN PREP AND COOLING DOWN MODES, D-1 SHALL OPEN. MAU-1 FAN SHALL MODULATE AS SCHEDULED WHEN EXHAUST FAN IS ENERGIZED.

3. WHEN MAKEUP AIR TEMPERATURE AT T-9 IS BELOW 50 DEGREES F, MANUFACTURER'S SUPPLIED CONTROLS SHALL OPEN VALVE V-1 TO MAINTAIN A DISCHARGE TEMPERATURE AT T-9. T-9 SHALL ALARM AT BAS WHEN TEMPERATURE DROPS BELOW 50 DEGREES F.

4. PREP AND COLING MODES: KITCHEN EXHAUST FANS SHALL MODULATE AT LOWEST SPEED AS POSSIBLE. MAU-1 IS DE-ENERGIZE.

INPUT / OUTPUT SUMMARY

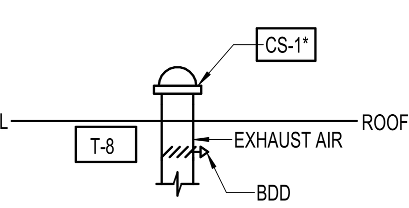
SYSTEM		INDICATION											ALARM				CONTROL					REMARKS
KITCHEN MAKEUP AIR UNIT AND GREASE EXHAUST FAN	GRAPHIC	STATUS	TEMPERATURE	RELATIVE HUMIDITY	PPM	CFM	STATIC PRESSURE	DIFFERENTIAL PRESSURE	GPM	POSITION	RUN TIME	HERTZ	LOW	HIGH	CRITICAL	MAINTENANCE	PROGRAM START /STOP	HAND-OFF-AUTO	MODULATE	MIN/MAX	OPEN/CLOSED	
SYSTEM	●	●																●				BAS, HOA
GREASE EXHAUST FAN		●									●	●				●			●			CS-2, VFD-2
MAKEUP AIR UNIT		●				●				●		●				●			●		●	CS-1, VFD-1, FS-1 , D-1
GAS HEAT			●										●						●			T-9,V-1
KITCHEN HOOD		●																				CNTL-1
HEATING COIL										●												T-9

BAS BUILDING AUTOMATION SYSTEM

T-8 TEMPERATURE SENSOR

HOA HAND-OFF-AUTOMATIC SWITCH

NOTES:
DEVICES NOTED WITH ARTWORK PROVIDE READ-OUT INFORMATION ONLY; NO CONTROL FUNCTION



ELEVATOR SHAFT VENTILATION

TEMPERATURE SENSOR, T-8, SHALL ENERGIZE FAN WHEN TEMPERATURE EXCEEDS 85 DEG. F (ADJ.).

INPUT / OUTPUT SUMMARY

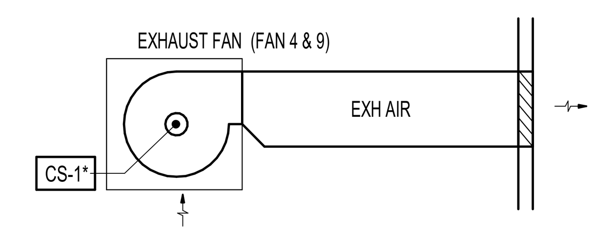
SYSTEM		INDICATION											ALARM				CONTROL					REMARKS
ROOF MOUNTED EXHAUST FAN	GRAPHIC	STATUS	TEMPERATURE	RELATIVE HUMIDITY	PPM	CFM	STATIC PRESSURE	DIFFERENTIAL PRESSURE	GPM	POSITION	RUN TIME	HERTZ	LOW	HIGH	CRITICAL	MAINTENANCE	PROGRAM START /STOP	HAND-OFF-AUTO	MODULATE	MIN/MAX	OPEN/CLOSED	
FAN-5, 6, & 7	●	●														●						BAS, HOA, CS-1*

BAS BUILDING AUTOMATION SYSTEM

HOA HAND-OFF-AUTOMATIC SWITCH

OS/ LS OCCUPANCY SENSOR (FAN-9)

T-7 SPACE MOUNTED TEMPERATURE SENSOR (FAN-4)



CEILING MOUNTED EXHAUST FAN

1. FAN SHALL BE STARTED AND STOPPED THROUGH A SYSTEM HAND-OFF-AUTOMATIC SWITCH. WHEN INDEXED TO "HAND", THE SYSTEM SHALL BE ENERGIZED TO OPERATE UNDER THE CONTROL SEQUENCE. WHEN INDEXED TO "OFF" THE SYSTEM SHALL BE DE-ENERGIZED. WHEN ENERGIZED TO "AUTOMATIC", THE SYSTEM SHALL BE STARTED AND STOPPED THROUGH THE BUILDING AUTOMATION SYSTEM.

2. ALL SAFETIES AND ASSOCIATED CONTROL INTERLOCKS SHALL REMAIN ACTIVE IN ALL H-O-A SWITCH POSITIONS.

3. SYSTEM IN AUTOMATIC POSITION:

FAN-4: SPACE TEMPERATURE SENSOR ,T-7, SHALL ENERGIZE FAN WHEN TEMPERATURE EXCEEDS 85 DEG. F (ADJ).

FAN-9: OCCUPANCY SENSOR, OS, SHALL START AND STOP FAN. WHEN FAN OS IS SWITCHED OFF, FAN SHALL CONTINUE TO OPERATE 10 MINUTES (ADJ.) BEFORE SHUTTING DOWN.

INPUT / OUTPUT SUMMARY

SYSTEM		INDICATION											ALARM				CONTROL					REMARKS
CEILING MOUNTED EXHAUST FAN	GRAPHIC	STATUS	TEMPERATURE	RELATIVE HUMIDITY	PPM	CFM	STATIC PRESSURE	DIFFERENTIAL PRESSURE	GPM	POSITION	RUN TIME	HERTZ	LOW	HIGH	CRITICAL	MAINTENANCE	PROGRAM START /STOP	HAND-OFF-AUTO	MODULATE	MIN/MAX	OPEN/CLOSED	
FAN-4 & 9	●	●														●						BAS, CS-1*

DEVICES NOTED WITH ASTERISK PROVIDE READ-OUT INFORMATION ONLY; NO CONTROL FUNCTION

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AUTOMATIC CONTROLS

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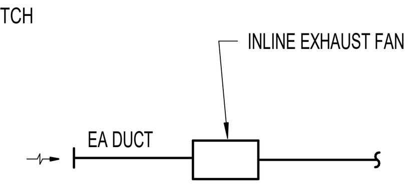
JOB N0.:201907.1

SHEET NO.
M702

BAS BUILDING AUTOMATION SYSTEM

HOA HAND-OFF-AUTOMATIC SWITCH



INLINE RESTROOM FAN

1. FAN SHALL BE SCHEDULED TO OPERATE THRU BAS.

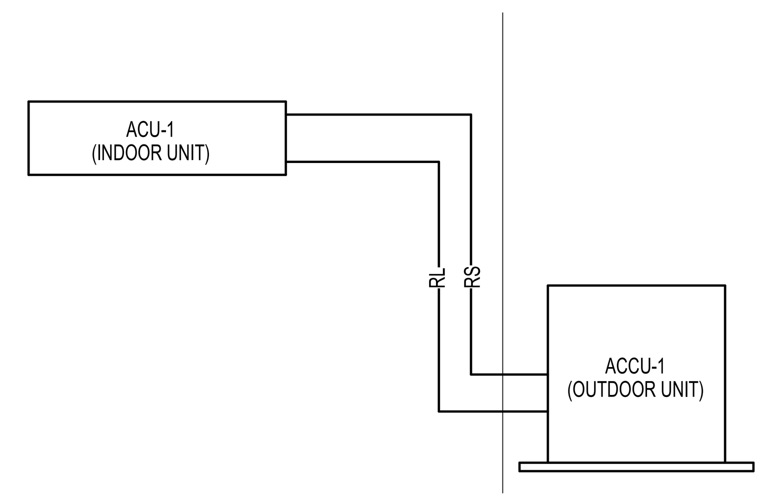
2. IF FAN FAILS TO START, AN ALARM SHALL BE SENT TO THE BAS.

INPUT/OUTPUT SUMMARY

EXHAUST FAN		INDICATION						ALARM				CONTROL					REMARKS
	GRAPHIC	STATUS	TEMPERATURE	CFM	DIFFERENTIAL PRESSURE	POSITION	HERTZ	STATUS	HIGH	CRITICAL	MAINTENANCE	PROGRAM S/S	ON-OFF-AUTO	MODULATE	MAX / MIN	OPEN / CLOSE	
FAN 2 & 3	●	●						●			●						

T-10 SPACE TEMPERATURE SENSOR

BAS BUILDING AUTOMATION SYSTEM



SPLIT SYSTEM (COOLING ONLY)

1. SPACE TEMPERATURE SENSOR, T-10, SHALL ENERGIZE DUCTLESS SPLIT SYSTEM WHEN SPACE TEMPERATURE RISES ABOVE 78 DEGREES IN COOLING MODE (ADJUSTABLE).

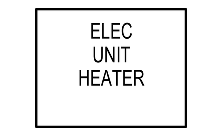
2. IF SPACE TEMPERATURE T-10 RISES ABOVE 95 DEGREES F, AN ALARM SHALL BE SENT TO THE BAS.

INPUT / OUTPUT SUMMARY

SYSTEM		INDICATION											ALARM				CONTROL					REMARKS
DUCTLESS SPLIT SYSTEM	GRAPHIC	STATUS	TEMPERATURE	RELATIVE HUMIDITY	PPM	CFM	STATIC PRESSURE	DIFFERENTIAL PRESSURE	GPM	POSITION	RUN TIME	HERTZ	LOW	HIGH	CRITICAL	MAINTENANCE	PROGRAM START /STOP	HAND-OFF-AUTO	MODULATE	MIN/MAX	OPEN/CLOSED	
SYSTEM	●																					BAS
ACCU		●																				CS-2*
ACU		●	●											●	●							CS-1*, T-1

T-11 UNIT MOUNTED TEMPERATURE SENSOR

BAS BUILDING AUTOMATION SYSTEM



UNIT HEATER

1. UNIT MOUNTED TEMPERATURE SENSOR, T-11, SHALL INITIATE FAN AND MODULATE ELECTRIC COIL TO MAINTAIN 65 DEGREES F SPACE TEMPERATURE (ADJUSTABLE).

2. WHEN T-11 IS SATISFIED. UNIT HEATER SHALL DE-ENERGIZE.

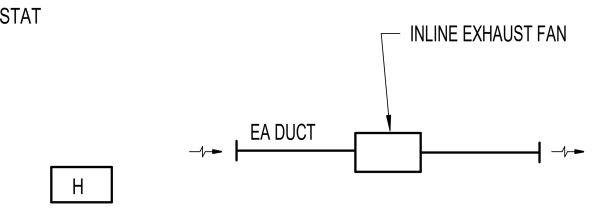
3. IF T-11 IS BELOW 40 DEG. F, AN ALARM SHALL BE SENT TO THE BAS.

INPUT / OUTPUT SUMMARY

SYSTEM		INDICATION											ALARM				CONTROL					REMARKS
UNIT HEATER	GRAPHIC	STATUS	TEMPERATURE	RELATIVE HUMIDITY	PPM	CFM	STATIC PRESSURE	DIFFERENTIAL PRESSURE	GPM	POSITION	RUN TIME	HERTZ	LOW	HIGH	CRITICAL	MAINTENANCE	PROGRAM START /STOP	HAND-OFF-AUTO	MODULATE	MIN/MAX	OPEN/CLOSED	
SYSTEM	●	●																				BAS, CS-1*
SPACE TEMP			●										●									T-1

BAS BUILDING AUTOMATION SYSTEM

H HUMIDISTAT



CRAWLSPACE FAN

1. FAN SHALL ENERGIZE WHEN HUMIDITY IS ABOVE 60% RH.

INPUT/OUTPUT SUMMARY

EXHAUST FAN		INDICATION						ALARM				CONTROL					REMARKS
	GRAPHIC	STATUS	TEMPERATURE	CFM	DIFFERENTIAL PRESSURE	POSITION	HERTZ	STATUS	HIGH	CRITICAL	MAINTENANCE	PROGRAM S/S	ON-OFF-AUTO	MODULATE	MAX / MIN	OPEN / CLOSE	
FAN 8	●	●						●			●						

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SHEET NO.

M703

PACKAGED VAV RTU WITH ENERGY RECOVERY WHEEL

UNIT NO	LOCATION	SERVICE	SUPPLY AIR		OUTDOOR AIR		SUPPLY FAN SECTION						EXHAUST FAN SECTION						FILTERS		HEATPUMP			AUXILIARY HEAT (NOTE 1)					DX COOLING COIL (NOTE 1)										ELECTRICAL		BASIS OF DESIGN	NOTES
			MAX	MIN	MAX	MIN	NO FANS	ESP	TSP	FLA	MAX BHP	HP	NO. FANS	ESP	TSP	FLA	MAX BHP	HP	PRE-FILTER MERV	FINAL FILTER MERV	HEATING CAP (MBH)	EAT DB	LAT (DB/WB)	MBH	AIR DATA EDB	AIR DATA LDB	INPUT CFH	INCOMING GAS PRESURE	TOTAL MBH	SENSIBLE MBH	EER	HEAT WHEEL CAP MBH	EDB	EWB	LDB	LWB	MAX COIL FV	APD	V	PH		
RTU-1	ROOF	1ST FLOOR	13500	4500	5000	1500	1	2.5	3.9	59.4	15.4	20	2	0.6	1.3	16.7	2.1	3 ea.	8	13	306	60	79	432	60.1	85.8	216	11"-13" w.c.	569	404	9.8	161	78	65	50	49.6	400	0.4	208	1	AAON RN050	2, 3
RTU-2	ROOF	2ND FLOOR	15600	8700	6300	3000	1	1.5	3.3	30.8	15.7	20	2	1.0	1.9	16.7	2.6	3 ea.	8	13	305	60	76	432	59.7	89.4	216	11"-13" w.c.	585	435	9.5	205	78	65	52	51.5	400	0.4	208	1	AAON RN050	2, 3

NOTES FOR PACKAGED VAV RTU WITH ENERGY RECOVERY WHEEL:

1. COIL SIZE IS DETERMINED BY RTU SIZE.
2. PROVIDE PROPANE GAS PRESSURE REGULATOR WITH UNIT TO REDUCE PROPANE GAS PRESSURE TO REQUIRED OPERATING PRESSURE.
3. MOUNT UNIT ON MANUFACTURER SUPPLIED PLENUM CURB.

GENERAL NOTES: (APPLICABLE TO ALL MECHANICAL SCHEDULES)

1. UNIT NUMBERS ARE INDICATED WHERE ALL UNITS ARE LISTED AND NUMBERED INDIVIDUALLY.
2. UNIT SYMBOLS ARE INDICATED WHERE ALL UNITS ARE NOT LISTED AND NUMBERED INDIVIDUALLY.
3. UNIT TYPES ARE DESCRIBED IN THE SPECIFICATIONS.
4. TEMPERATURE VALUES ARE LISTED IN DEGREES FAHRENHEIT.
5. AIR PRESSURE VALUES ARE LISTED IN INCHES OF WATER COLUMN.
6. AIR VELOCITY VALUES ARE LISTED IN FEET PER MINUTE.
7. DUCT AND PIPE SIZES ARE LISTED IN SINGLE-NUMBER INCHES OF NOMINAL DIAMETER OR MULTIPLE NUMBER INCHES OF INDICATED PARAMETER.CONNECTION SIZES ARE BRANCH SIZES FROM MAINS TO UNIT INLETS. SINGLE PIPE SIZE IS TYPICAL FOR SUPPLY AND RETURN.

AIR BALANCE (ENTIRE BUILDING HOODS ON)

UNIT NO.	OA (CFM)	EA (CFM)	BALANCE (CFM)
FAN-1	-	2065	-2065
FAN-2	-	200	-200
KEF-1	-	1645	-1645
KEF-2	-	1175	-1175
KEF-3	-	2400	-2400
KEF-4	-	725	-725
MAU-1	3816	-	+3816
RTU-2	6300	-	+6300
TOTAL AIRFLOW	15,116	9,010	+6,106

NOTE: BUILDING IS POSITIVELY PRESSURED 6,106 CFM. RELIEVED THROUGH RELIEF FAN TO MAINTAIN 0.05" W.G. BUILDING STATIC PRESSURE.

AIR BALANCE (ENTIRE BUILDING HOODS OFF)

UNIT NO.	OA (CFM)	EA (CFM)	BALANCE (CFM)
FAN-1	-	2065	-2065
FAN-2	-	200	-200
KEF-1	-	OFF	0
KEF-2	-	OFF	0
KEF-3	-	OFF	0
KEF-4	-	OFF	0
MAU-1	OFF	-	0
RTU-2	6300	-	+6300
TOTAL AIRFLOW	11,300	3,365	+7,935

NOTE: BUILDING IS POSITIVELY PRESSURED 7,935 CFM. RELIEVED THROUGH RELIEF FAN TO MAINTAIN 0.05" W.G. BUILDING STATIC PRESSURE.

AIR BALANCE (CBQ HOODS ON)

UNIT NO.	OA (CFM)	EA (CFM)	BALANCE (CFM)
MAU-1	3816	-	+3816
KEF-1	-	1645	-1645
KEF-2	-	1175	-1175
KEF-3	-	2400	-2400
KEF-4	-	725	-725
TOTAL AIRFLOW	3,816	5,945	-2,129

NOTE: KITCHEN IS NEGATIVELY PRESSURED 2,129 CFM.

AIR BALANCE (CBQ HOODS OFF)

UNIT NO.	OA (CFM)	EA (CFM)	BALANCE (CFM)
MAU-1	OFF	-	0
KEF-1	-	OFF	0
KEF-2	-	OFF	0
KEF-3	-	OFF	0
KEF-4	-	OFF	0
TOTAL AIRFLOW	0	0	0

NOTE: KITCHEN IS NOT PRESSURED.

FANS

UNIT NO.	TYPE	LOCATION	SERVICE	CFM	ESP	FAN RPM	MOTOR HP	MOTOR W	ELECTRICAL V	ELECTRICAL PH	BASIS OF DESIGN	NOTES
FAN-1	I	ROOF	1ST & 2ND FLRS. RR	2065	0.5	1725	3/4	-	208	3	GREENHECK G-133-A	
FAN-2	P	STORAGE - 125	BAIT & TACKLE RR	200	0.5	1552	1/10	-	120	1	GREENHECK SQ-80-VG	
FAN-3	P	STORAGE 306	3RD FLR RR	740	0.5	1624	1/4	-	120	1	GREENHECK SQ-99-VG	
FAN-4	L	POOL EQUIP - 307	POOL EQUIP - 307	360	0.5	1275	-	155	120	1	GREENHECK SP-A510-VG	
FAN-8	E	CRAWL SPACE	CRAWL SPACE VENTILATION	1000	0.25	1770	1/3	-	120	1	GREENHECK AX-31-160-0627-A3	
FAN-9	L	OUTDOOR GAMING - 222	OUTDOOR GAMING - 222	500	0.25	1275	-	155	120	1	GREENHECK SP-A510-VG	
FAN-10	I	ELEC RM 115 ROOF	ELEC RM 115	300	0.5	1666	1/10	-	120	1	GREENHECK G-080-VG	
~~FAN-11~~	~~L~~	~~ELEC RM 210~~	~~ELEC RM 210~~	~~150~~	~~0.25~~	~~1400~~	~~-~~	~~113~~	~~120~~	~~1~~	~~GREENHECK SP-A190~~	
FAN-12	I	ROOF	SMOKING ROOM	300	0.5	1300	1/12	-	120	1	GREENHECK G-095-G	

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DUCTLESS SPLIT AIR CONDITIONING UNITS

UNIT NO INDOOR	UNIT NO OUTDOOR	SERVICE	FAN CFM	TOT MBH	EAT DB	EAT WB	ELECTRICAL V	ELECTRICAL PH	BASIS OF DESIGN INDOOR	BASIS OF DESIGN OUTDOOR	NOTES
ACU-1	ACCU-1	SERVER - 211	705	24	80	67	208	1	MITSUBISHI PKA-A24KA	MITSUBISHI PUY-A24NHA	

NOTES FOR DUCTLESS SPLIT AIR CONDITIONERS:
1. UNIT IS COOLING ONLY.

GAS-FIRED RADIANT HEATERS

UNIT NO.	LOCATION	APPROX UNIT SIZE (IN) W	L	D	INPUT MBH	PIPE SIZE	ELECTRICAL V	ELECTRICAL PH	AMPS	BASIS OF DESIGN	NOTES
RH-1	OUTDOOR GAMING - 222	33	302	12	160	3/4	120	1	1	VANTAGE TF-160LPG	PROPANE GAS

UNIT HEATERS

UNIT NO.	LOCATION	APPROX UNIT SIZE (IN) W	H	D	CFM	KW	ELECTRICAL V	ELECTRICAL PH	AMPS	BASIS OF DESIGN	NOTES
UH-1.1	VEST - 100	16	20	6	100	1.5	120	1	12.5	QMARK AWH3150F	1
UH-1.2	ELEC - 115	14	13	20	270	2.5	208	1	12	QMARK MWUH5004	1, 2
UH-1.3	STAIRS - 110	16	20	6	100	4.8	208	3	13.3	QMARK AWH45083F	1
UH-1.4	VEST - 121	16	20	6	100	1.5	120	1	12.5	QMARK AWH3150F	1
UH-1.5	STAIRS - 120	16	20	6	100	4.8	208	3	13.3	QMARK AWH45083F	1
UH-1.6	WATER - 116	14	13	20	270	2.5	208	1	12.0	QMARK MWUH5004	1, 2
UH-2.1	SERVICE CORR - 218	16	20	6	100	2	208	1	9.6	QMARK AWH3180F	1
~~UH-2.2~~	~~VEST - 221~~	~~16~~	~~20~~	~~6~~	~~100~~	~~1.5~~	~~120~~	~~1~~	~~12.5~~	~~QMARK AWH3150F~~	~~1~~
UH-2.3	SMOKING ROOM	16	20	6	100	2	208	1	9.6	QMARK AWH3180F	1
~~UH-3.1~~	~~STORAGE - 306~~	~~14~~	~~13~~	~~20~~	~~270~~	~~2.5~~	~~208~~	~~1~~	~~12.0~~	~~QMARK MWUH5004~~	~~1, 2~~
UH-3.2	POOL EQUIP - 307	14	13	20	270	2.5	208	1	12.0	QMARK MWUH5004	1, 2
UH-3.3	POOL DECK CORRIDOR	16	20	6	100	2	208	1	9.6	QMARK AWH3180F	1
UH-3.4	MEN - 304	16	20	6	100	2	208	1	9.6	QMARK AWH3180F	1
UH-3.5	WOMEN - 305	16	20	6	100	2	208	1	9.6	QMARK AWH3180F	1

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NOTES FOR UNIT HEATERS
1. INTEGRAL THERMOSTAT
2. FURNISH WITH WALL BRACKET

KITCHEN EXHAUST FAN

UNIT NO.	TYPE	LOCATION	SERVICE	CFM	ESP	FAN RPM	MOTOR HP	ELECTRICAL V	ELECTRICAL PH	BASIS OF DESIGN	NOTES
KEF-1	J2	ROOF	HOOD 1 (LEFT)	1645	1.0	959	1.0	208	3	CAPTIVEAIRE DU180HFA	SEE KITCHEN CONSULTANT'S DRAWINGS
KEF-2	J2	ROOF	HOOD 1 (RIGHT)	1175	1.0	937	1.0	208	3	CAPTIVEAIRE DU180HFA	SEE KITCHEN CONSULTANT'S DRAWINGS
KEF-3	J2	ROOF	HOOD 3	2400	2.0	1396	3.0	208	3	CAPTIVEAIRE CASRE20DD	SEE KITCHEN CONSULTANT'S DRAWINGS
KEF-4	J2	ROOF	HOOD 4	725	0.5	1345	1/3	120	1	CAPTIVEAIRE DU33HFA	SEE KITCHEN CONSULTANT'S DRAWINGS

NOTE: EQUIPMENT ABOVE IS FOR INFORMATION ONLY. EQUIPMENT FURNISHED UNDER ANOTHER DIVISION AND INSTALLED UNDER DIVISION 23.

MAKEUP AIR UNIT

UNIT NO	LOCATION	SERVICE	SUPPLY FAN CFM	SUPPLY FAN ESP	SUPPLY FAN MOTOR HP	GAS REHEAT TEMP RISE	GAS REHEAT INPUT MBH	GAS REHEAT OUTPUT MBH	GAS REHEAT INCOMING GAS PRESS	GAS REHEAT PIPE SIZE	ELECTRICAL	BASIS OF DESIGN	NOTES
MAU-1	ROOF	KITCHEN	3816	1.0	5	57	252	232	STD	1	208V/3PH	CAPTIVEAIRE A2-D.500-20D	SEE KTICHEN CONSULTANT'S DRAWINGS

NOTE: EQUIPMENT ABOVE IS FOR INFORMATION ONLY. EQUIPMENT FURNISHED UNDER ANOTHER DIVISION AND INSTALLED UNDER DIVISION 23.

SPLIT SYSTEM HEAT PUMPS (NOTES 3,4,6)

UNIT NO.	OUTDOOR	LOCATION/SERVICE	FAN CFM	FAN WATTS	FAN ESP	OA CFM	COOLING CAPACITY (NOTE 1) TOTAL MBH	COOLING CAPACITY (NOTE 1) SENS. MBH	EFFICIENCY SEER	HEATING CAPACITY (NOTE 2) REVERSE CYCLE MBH	HEATING CAPACITY (NOTE 2) SUPPLEMENTAL KW	HEATING CAPACITY (NOTE 2) HSPF	ELECTRICAL VOLTAGE	ELECTRICAL PHASE	ELECTRICAL HERTZ	BASIS OF DESIGN (NOTE 5) INDOOR UNIT	BASIS OF DESIGN (NOTE 5) OUTDOOR UNIT	NOTES
FCU-1	HPU-1	ELEVATOR SHAFT #1	450	85	0.3	-	12.0	10.1	18.6	16.3	-	10.9	208	1	60	MITSUBISHI PEAD-A12AA7	MITSUBISHI SUZ-KA12NAR1	
FCU-2	HPU-2	ELEVATOR SHAFT #2	450	85	0.3	-	12.0	10.1	18.6	16.3	-	10.9	208	1	60	MITSUBISHI PEAD-A12AA7	MITSUBISHI SUZ-KA12NAR1	
FCU-3	HPU-3	ELEVATOR SHAFT #3	450	85	0.3	-	12.0	10.1	18.6	16.3	-	10.9	208	1	60	MITSUBISHI PEAD-A12AA7	MITSUBISHI SUZ-KA12NAR1	

NOTES FOR SPLIT SYSTEM HEAT PUMPS:

1. COOLING CAPACITY BASED ON 95 DEGREES F DB CONDENSER ENTERING AIR TEMPERATURE AND 80 DEGREES F DB/67 DEGREES F WB INDOOR CONDITIONS.
2. HEATNG CAPCITY BASED ON 47 DEGREES F OUTDOOR AIR TEMPERATURE AND 70 DEGREES F ROOM AIR TEMPERATURE ENTERING THE COIL.
3. PROVIDE INDOOR UNIT WITH INTEGRAL CONDENSATE PUMP/LIFT MECHANISIM.
4. COOLING/ HEATING CAPACITY INDICATED IS BASIS OF DESIGN MANUFACTURER RATING. SUBMIT HEATING AND COOLING CAPACITY TO ENGINEER FOR REVIEW AND APPROVAL PRIOR TO EQUIPMENT PURCHASE. CAPACITIES SHALL BE DE-RATED FOR INDOOR AND OUTDOOR AIR TEMPERATURES AND EQUIVILENT REFRIGERANT LINE LENGTHS. INCLUDE DE-RATED HEATING CAPACITY AT 0 DEGREES F IN SUBMITTAL.
5. R-410A REFRIGERANT, SIZE AND PROVIDE LONG LINE ACCESSORIES IN ACCORDANCE WITH MANUFACTURER'S REQUIREMENTS AND RECOMMENDATIONS FOR LONG LINE APPLICATION. CONTRACTOR SHALL VERIFY FINAL REQUIRED REFRIGERANT (LBS) BASED ON FIELD CONDITIONS AND ACTUAL PIPING LENGTH.
6. REFRIGERANT CIRCUIT ACCESS PORTS LOCATED OUTDOORS SHALL BE FITTED WITH LOCKING-TYPE TAMPERRESISTANT CAPS OR SHALL BE OTHERWISE SECURED TO PREVENT UNAUTHORIZED ACCESS.

SINGLE DUCT AIR TERMINAL UNITS, ELECTRIC REHEAT

UNIT NO.	TYPE	MAX APD	MAX CFM	MIN CFM	HEATING CFM	BRANCH DUCT SIZE	BOX SIZE	ELECTRIC REHEAT EAT	ELECTRIC REHEAT LAT	ELECTRIC REHEAT kW	ELECTRIC REHEAT CONTROL	V/ PH	NOTES
ATU-1.1	B	0.5	800	200	200	14	12	55	95	2.5	SCR	208/1	
ATU-1.2	B	0.5	1500	350	350	16	14	55	95	4.5	SCR	208/3	
ATU-1.3	B	0.5	1500	350	350	16	14	55	95	4.5	SCR	208/3	
ATU-1.4	B	0.5	1500	150	150	16	14	55	95	2.0	SCR	208/1	
ATU-1.5	B	0.5	1500	150	150	16	14	55	95	2.0	SCR	208/1	
ATU-1.6	B	0.5	1300	500	450	16	14	55	76	3.0	SCR	208/1	
ATU-1.7	B	0.5	1400	450	450	16	14	55	76	3.0	SCR	208/1	
ATU-1.8	B	0.5	1500	500	450	16	14	55	76	3.0	SCR	208/1	
ATU-1.9	B	0.5	1200	450	450	12	10	55	76	3.0	SCR	208/1	
ATU-1.10	B	0.5	1200	450	450	12	10	55	76	3.0	SCR	208/1	
ATU-1.11	B	0.5	1200	450	450	12	10	55	76	3.0	SCR	208/1	
ATU-1.12	B	0.5	300	100	100	10	8	55	95	1.0	SCR	120/1	
ATU-1.13	B	0.5	1000	300	300	12	10	55	95	4.0	SCR	208/3	
ATU-1.14	B	0.5	300	100	100	10	8	55	95	1.0	SCR	120/1	
ATU-2.1	B	0.5	800	200	200	12	10	55	95	2.5	SCR	208/1	
ATU-2.2	B	0.5	1200	800	800	12	10	55	71	4.0	SCR	208/3	
ATU-2.3	B	0.5	700	400	400	10	8	55	75	2.5	SCR	208/1	
ATU-2.4	B	0.5	700	400	400	10	8	55	75	2.5	SCR	208/1	
ATU-2.5	B	0.5	1400	800	800	16	14	55	71	4.0	SCR	208/3	
ATU-2.6	B	0.5	1400	800	800	16	14	55	71	4.0	SCR	208/3	
ATU-2.7	B	0.5	1500	800	800	16	14	55	71	4.0	SCR	208/3	
ATU-2.8	B	0.5	1100	800	800	12	10	55	71	4.0	SCR	208/3	
ATU-2.9	B	0.5	1400	800	800	16	14	55	71	4.0	SCR	208/3	
ATU-2.10	B	0.5	700	200	300	10	8	55	97	4.0	SCR	208/3	
ATU-2.11	B	0.5	1400	800	800	16	14	55	71	4.0	SCR	208/3	
ATU-2.12	B	0.5	300	100	100	8	6	55	87	1.0	SCR	120/1	
ATU-2.13	B	0.5	1200	800	800	12	10	55	71	4.0	SCR	208/3	
ATU-2.14	B	0.5	300	100	100	8	6	55	87	1.0	SCR	120/1	
ATU-2.15	B	0.5	1400	800	800	16	14	55	71	4.0	SCR	208/3	
ATU-3.1	B	0.5	1200	200	450	12	10	55	71	4.0	SCR	208/3	

NOTES FOR SINGLE DUCT TERMINAL UNITS:
1. BRANCH DUCT SIZE IS RUNOUT AND NOT NECESSARILY THE BOX INLET SIZE. PROVIDE 3 DUCT DIAMTERS OF STRAIGHT DUCT ON BOX INLET WITH THE LAST 1 FT OF DUCT THE SAME SIZE AS BOX INLET. PROVIDE INSULATED TRANSITION IF REQUIRED FOR CONNECTION.

2. CONTRACTOR RESPONSIBLE FOR COORDINATING RIGHT AND LEFT HAND ORIENTATION.

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SEAL:

CERTIFICATION:
I certify that these documents were prepared or approved by me, and that I am a duly licensed engineer under the laws of the State of Maryland,
license number 25187, expiration date 01/15/22.

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PROJECT:

ROD N' REEL RESORT

4160 MEARS AVE.
CHESAPEAKE BEACH, MD 20732

REVISIONS:

01/15/2021	FOR PERMIT
04/15/2021	PERMIT RESPONSE
05/15/2021	BID SUBMISSION
06/04/2021	ADDENDUM 1
07/14/2021	ADDENDUM 2
10/15/2021	CONSTRUCTION CONFORMING DOCUMENTS
07/22/2022	CHANGE BULLETIN 5
09/15/2022	CHANGE BULLETIN 6

TITLE:
MECHANICAL SCHEDULES

DATE: 10.15.2021
JOB NO.:201907.1

SHEET NO.
M801